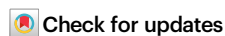


Allelic variation of an LRR only protein confers dual pathogen resistance in *Arabidopsis*

Received: 25 November 2025

Accepted: 19 June 2026

Published online: 07 July 2026



Bao Liu^{1,2,3}, Yudan Zheng^{1,2,3}, Shangling Lou¹, Jing Hou¹, Xiaoqin Feng¹, Landi Feng¹, Yan Song¹, Dong Zhang¹, Yuelin Zhang¹, Yuanzhong Jiang¹✉ & Jianquan Liu^{1,2}✉

Plants utilize versatile leucine-rich repeat (LRR) proteins to perceive diverse pathogen-derived signals, thereby activating or suppressing defense responses. However, the specific roles of LRR-only proteins in pathogen-specific resistance remain largely unclear. Here, we show that allelic variations in an *Arabidopsis thaliana* LRR-only gene, *LEUCINE-RICH REPEAT IMMUNE PROTEIN 1* (*LRIPI*), underlies dual functions in combating distinct pathogens. An allele common in arid regions confers resistance to *Pseudomonas syringae* pv. *tomato* (*Pst*) bacteria via salicylic acid (SA) signaling pathway, whereas an allele enriched in high-precipitation regions protects against *Botrytis cinerea* fungus via camalexin biosynthesis. Divergence in promoter and coding sequences between the two alleles modulates gene expression and elicitor-associated immune responses, thereby promoting pathogen-specific complex formation with BRASSINOSTEROID INSENSITIVE 1 (BRI1)-ASSOCIATED RECEPTOR KINASE 1 (BAK1). Globally distributed *A. thaliana* accessions carry two distinct *LRIPI* allele groups that correlate with local precipitation conditions. These findings demonstrate that a single LRR-only gene can evolve environmentally adapted alleles that optimize immunity against different pathogens in contrasting habitats.

Plants coexist with a diverse array of microbial pathogens, necessitating sophisticated immune mechanisms to balance growth and defense^{1–4}. Central to plant immunity are two interconnected layers: pattern-triggered immunity (PTI) and effector-triggered immunity (ETI)^{5,6}. PTI is activated when cell-surface pattern recognition receptors (PRRs) detect conserved pathogen-associated molecular patterns (PAMPs), triggering broad-spectrum defenses, such as reactive oxygen species (ROS) bursts and defense gene expression^{6,7}. ETI, mediated by intracellular nucleotide-binding leucine-rich repeat (NB-LRR) receptors that recognize pathogen effectors^{8,9}, amplifies these responses,

often culminating in localized cell death. While PTI and ETI were historically viewed as distinct pathways, emerging evidence highlights their synergy, with PTI priming ETI and shared signaling components like respiratory burst oxidase homolog D (RBOHD) integrating ROS production across both pathways^{10,11}. For instance, RBOHD activation depends on PRR-mediated phosphorylation during ETI, underscoring PTI's foundational role in immune amplification¹². This crosstalk is further modulated by lipid signaling molecules like phosphatidic acid (PA), stabilized by diacylglycerol kinase 5, and kinase networks involving botrytis-induced kinase 1 and mitogen-activated protein kinase 4,

¹Sichuan Zoige Alpine Wetland Ecosystem National Observation and Research Station, Key Laboratory for Bio-resources and Eco-environment & State Key Lab of Hydraulics & Mountain River Engineering, College of Life Sciences, Sichuan University, Chengdu, China. ²State Key Laboratory of Herbage Improvement and Grassland Agro-ecosystems, College of Ecology, Lanzhou University, Lanzhou, China. ³These authors contributed equally: Bao Liu, Yudan Zheng.

✉ e-mail: liujq@nwpb.cas.cn; jyz88623@126.com

which fine-tune immune outputs by balancing PA production and degradation¹³. Additionally, convergence points, such as ADRI-family NLRs interacting with enhanced disease susceptibility 1 and phytoalexin deficient 4, exemplify the molecular integration of PTI and ETI, enabling plants to mount context-dependent defenses¹⁴.

The LRR superfamily is defined by tandem arrays of LRR motifs, typically 20–30 amino acids in length and characterized by a conserved consensus sequence enriched in leucine residues¹⁵. These motifs form structural scaffolds that mediate protein–protein interactions and contribute to ligand perception in plant–microbe interactions. Based on domain architecture, cell-surface LRR-containing proteins can be broadly classified into several groups. LRR receptor kinases (LRR-RKs), including the well-characterized *Arabidopsis* pattern-recognition receptors flagellin-sensing 2 (FLS2) and EF-Tu receptor, contain an extracellular LRR domain that recognizes the bacterial ligands flagellin and elongation factor-thermostable (EF-Tu), respectively, a single-pass transmembrane domain, and an intracellular kinase domain that activates downstream signaling upon ligand perception, typically together with co-receptors, such as brassinosteroid insensitive 1 (BRI1)-associated receptor kinase 1 (BAK1)^{16,17}. LRR receptor proteins (LRR-RPs) lack intracellular kinase domains and therefore rely on associated kinases for signal transduction. For example, receptor-like protein 23 (RLP23) recognizes the conserved necrosis and ethylene-inducing peptide 1-like protein-derived peptide nlp20^{18,19}, RLP32 detects the bacterial translation initiation factor IF1²⁰, and RLP42 senses fungal endopolysaccharidases²¹. These receptors constitutively associate with the adaptor kinase suppressor of BIR1-1 (SOBIR1), an LRR-RK-like protein, and recruit BAK1 upon ligand recognition to activate immune signaling^{18–21}. Intracellular NB-LRR proteins constitute the major class of plant resistance (R) proteins. Their NB domain functions as an ADP–ATP molecular switch that regulates receptor activation²², whereas the C-terminal LRR domain contributes to effector recognition and suppresses autoactivation in the absence of pathogen signals^{8,9,23}.

In addition to these canonical receptor classes, LRR-only proteins have been reported in several plant species and lack both transmembrane regions and recognizable intracellular signaling domains. For example, several LRR-only proteins negatively regulate immune responses by destabilizing receptor complexes or suppressing downstream signaling. *N. benthamiana* LRR protein 1 (NbLRRP1) attenuates sustained PTI through interaction with NbBAK1 and disruption of immune receptor complexes, including NbBAK1–NbFLS2 and NbBAK1–receptor-like protein (REO2)²⁴. Similarly, in *Nicotiana tabacum*, *NtLRP1* is induced early during the hypersensitive response, and its overexpression has been shown to suppress HR²⁵, while *CaLRR1* in *Capsicum annuum* inhibits cell death via interactions with membrane-associated proteins, such as hypersensitive induced reaction protein 1 (CaHIR1) and pathogenesis-related protein 4B (CaPR4b)²⁶. Conversely, other LRR-only proteins enhance immune signaling by promoting ligand perception or stabilizing immune-related complexes. An extracellular LRR-only protein has been shown to bind necrosis- and ethylene-inducing peptide-like proteins (NLPs) and promote their oligomerization, thereby potentiating immune activation²⁷. In rice, *OsLRR1* contributes to XA21-mediated resistance to *Xanthomonas oryzae*, as silencing of *LRR1* compromises resistance, and is associated with the localization of its interacting partner *OshIR1*^{28,29}. Despite these advances, whether and how LRR-only proteins can mediate pathogen-specific immune responses, particularly through allelic variation, remains unclear.

In this study, we found two *A. thaliana* ecotypes from arid and high-precipitation regions, exhibiting significantly different responses to contrasting infections by bacterial and fungal pathogens. Notably, our genome-wide association studies (GWAS) of the recombinant inbred line (RIL) population between them identified a single gene primarily responsible for this variation. We further characterized this

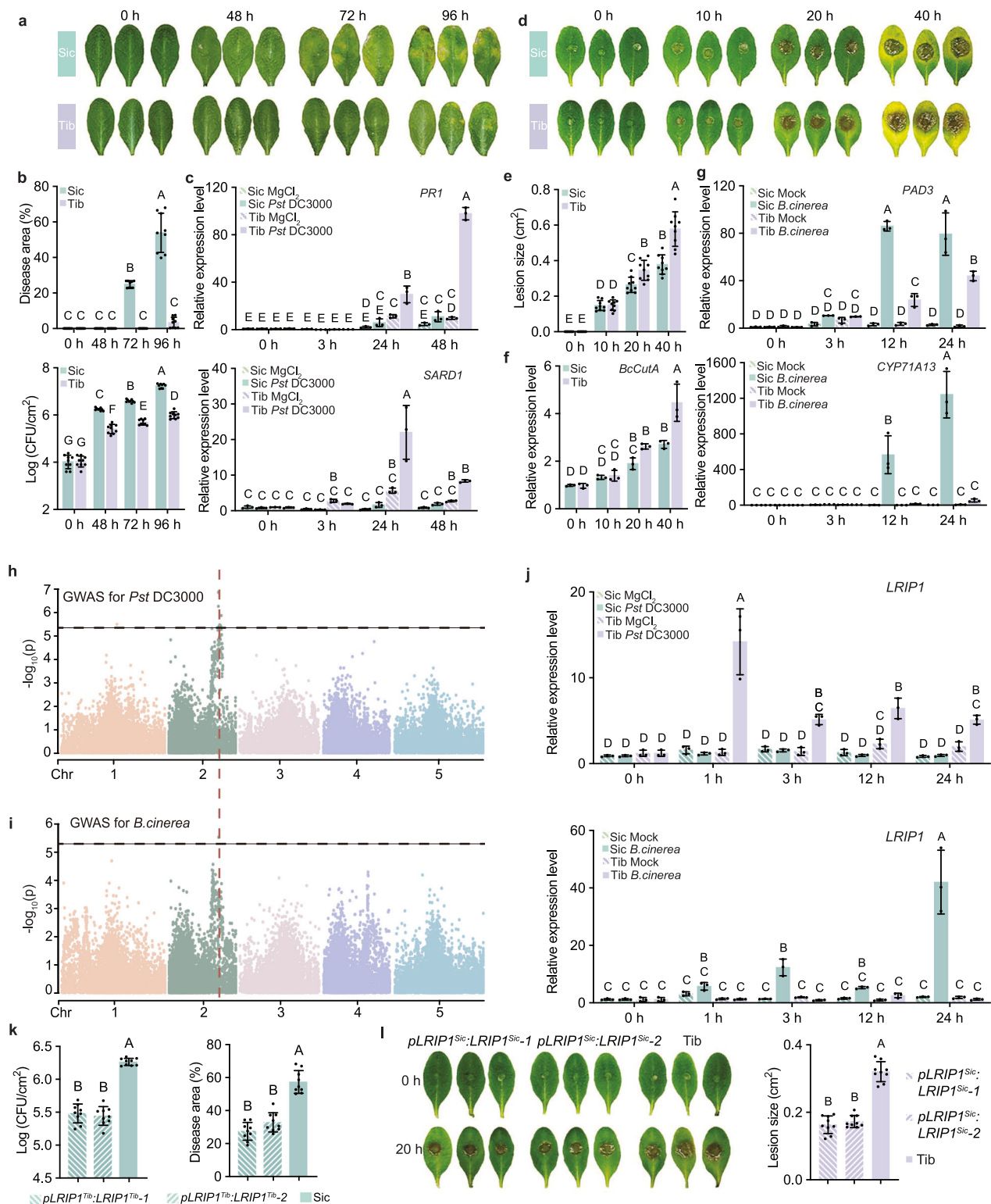
gene as a LRR-only protein, designated *receptor-like protein resistance 1* (*LRPI*). The allele from arid habitats confers resistance to the bacterial pathogen *Pseudomonas syringae* pv. *tomato* strain DC3000 (*Pst* DC3000) through salicylic acid (SA)-dependent pathways. Conversely, the allele from high-precipitation regions activates camalexin biosynthesis to combat the fungal pathogen *Botrytis cinerea*. Mechanistically, promoter polymorphisms drive pathogen-specific induction of *LRPI*, whereas variation in the LRR coding sequence (CDS) fine-tunes elicitor-associated immune responses. Both alleles constitutively interact with SOBIR1 but form pathogen-specific complexes with BAK1, revealing a dynamic co-receptor recruitment mechanism. Our findings illustrate how a single LRR-only gene can evolve functionally distinct alleles that optimize immunity against diverse pathogens.

Results

Allelic variations at the *AT2G34930* locus are linked to differential immune responses against *Pst* DC3000 and *B. cinerea*

We initially used two ecotypes of *A. thaliana*: the Tibet ecotype (Tib), originating from a high-altitude, arid habitat, and the Sichuan ecotype (Sic), found in a low-altitude, high-precipitation habitat, to explore whether they exhibit differential immune responses to different pathogens. Phylogenetic analysis revealed that these two ecotypes belong to distinct groups of the species³⁰. We exposed them to two pathogens with different lifestyles: the bacterium *Pst* DC3000, which thrives by extracting nutrients from living host cells, and the fungus *B. cinerea*, which promotes infection by inducing host cell death and exploiting necrotic tissue as a nutrient source^{31,32}. Resistance was assessed by quantifying bacterial colony numbers (*Pst* DC3000) and lesion areas (*B. cinerea*) as indicators of immune response. Following *Pst* DC3000 inoculation, Tib exhibited smaller disease areas, reduced disease symptoms, and significantly lower bacterial proliferation compared to Sic (Fig. 1a, b). Consistent with its resistance phenotype, Tib exhibited a significantly enhanced response to SA, along with elevated expression of SA pathway markers *pathogenesis-related gene 1* (*PR1*) and *SAR deficient 1* (*SARD1*) post-inoculation compared to Sic ecotype^{33,34} (Fig. 1c and Supplementary Fig. 1a, b), consistent with the canonical role of SA signaling in bacterial pathogen defense. In contrast, Sic showed pronounced resistance to *B. cinerea*^{35,36}, with significantly smaller lesions and reduced fungal biomass relative to Tib (Fig. 1d–f). This resistance correlated with upregulation of camalexin biosynthesis genes *phytoalexin deficient 3* (*PAD3*) and *CYP71A13*^{37,38} (Fig. 1g), consistent with the established role of this phytoalexin in fungal defense. Intriguingly, neither ecotype exhibited differential sensitivity to jasmonic acid (JA) nor altered expression of JA-responsive genes (Supplementary Fig. 1c–f), suggesting that their contrasting resistance mechanisms operate independently of SA–JA cross-regulation.

To identify genetic loci underlying these phenotypes, we performed GWAS on 120 recombinant inbred lines (RILs; F7 generation) derived from a Sic × Tib cross. A significant GWAS peak on chromosome 2 was associated with *Pst* DC3000 resistance (Fig. 1h and Supplementary Fig. 2a). Strikingly, parallel GWAS for *B. cinerea* resistance revealed a colocalizing peak (Fig. 1i and Supplementary Fig. 2b), implicating a shared genomic region. The most strongly associated SNP resided within *AT2G34930*, encoding a leucine-rich repeat protein which we designated as *leucine-rich repeat immune protein 1* (*LRPI*) (Supplementary Fig. 2b, d)^{39,40}. Expression profiling revealed pathogen-specific induction of *LRPI*: in Tib, *LRPI* was rapidly upregulated following *Pst* DC3000 inoculation but unresponsive to *B. cinerea*, while Sic exhibited sustained *LRPI* induction specifically during *B. cinerea* infection (Fig. 1j). Functional validation via reciprocal transformation demonstrated that introducing *pLRPI^{Tib}·LRPI^{Tib}* into Sic enhanced *Pst* DC3000 resistance (Fig. 1k and Supplementary Fig. 2e), whereas expressing *pLRPI^{Sic}·LRPI^{Sic}* in Tib attenuated *B. cinerea* susceptibility (Fig. 1l and Supplementary Fig. 2f). These results



indicate *LRIP1* as a critical determinant of pathogen-specific resistance, with allelic variation driving divergent defense prioritization in Tib and Sic.

Alleles of *LRIP1* confer differential resistance to fungal and bacterial pathogens

To dissect the allelic variation in *LRIP1* underlying its pathogen-specific functions in the Tib and Sic ecotypes, we analyzed single-nucleotide polymorphisms (SNPs) across the *LRIP1* locus, including the -2-kb promoter region and CDS. Comparative sequencing identified 150

SNPs in the promoter (66 substitutions, 10 indels; Supplementary Fig. 3a) and 59 SNPs in the coding region, including 31 non-synonymous substitutions (Fig. 2a). To assess the functional relevance of promoter polymorphisms, we generated transgenic *A. thaliana* Col lines expressing β -glucuronidase (GUS) under the control of either the *LRIP1*^{Tib} or *LRIP1*^{Sic} promoter. Basal *LRIP1* expression was negligible in uninoculated plants. However, *LRIP1*^{Tib} promoter activity was strongly induced upon *Pst* DC3000 infection, while *LRIP1*^{Sic} was strongly induced by *B. cinerea* (Supplementary Fig. 3b). This pathogen-specific transcriptional activation was corroborated by luciferase-

Fig. 1 | *LRP1* confers resistance to *Pst* DC3000 and *B. cinerea*. **a** Representative leaf images showing disease progression in Sic and Tib accessions following *Pst* DC3000 inoculation at 0, 48, 72, and 96 h post-inoculation. **b** Quantification of diseased leaf area relative to total leaf area (upper) and bacterial colony-forming units (CFUs; lower) in Sic and Tib accessions following *Pst* DC3000 inoculation. **c** Expression of *PR1* and *SARD1* in Sic and Tib accessions after *Pst* DC3000 or MgCl₂ treatment over time (0, 3, 24, and 48 h). **d** Lesion development in the Sic and Tib accessions following *B. cinerea* inoculation at 0, 10, 20, and 40 h post-inoculation. **e** Quantification of the lesion size in **(d)**. **f** mRNA levels of *BcCutA*, reflecting fungal growth, in Sic and Tib accessions over time. **g** Expression levels of *PAD3* and *CYP71A13* in Sic and Tib accessions following inoculation with *B. cinerea*. Equal-sized MS agar blocks were used as mock controls. GWAS results for resistance to *Pst* DC3000 (**h**) and *B. cinerea* (**i**). Manhattan plots indicate significant loci associated

with resistance to *Pst* DC3000 or *B. cinerea*. The significance threshold ($P < 0.05/n$, where n represents the number of tested SNPs) was determined using Bonferroni correction. **j** Expression of *LRP1* in Sic and Tib accessions after inoculation with *Pst* DC3000, MgCl₂, agar (mock), and *B. cinerea*. **k** Leaf image (left) and quantification of disease area percentages relative to the total leaf area (right) in *pLRP1^{Tib}:LRP1^{Tib}*-1/2 transgenic lines in the Sic background and Sic accession following 96 h post-inoculation with *Pst* DC3000. **l** Lesion development (left) and quantification of lesion size (right) in *pLRP1^{Sic}:LRP1^{Sic}*-1/2 transgenic lines in the Tib background and Tib accession following 20 h post-inoculation with *B. cinerea*. In **b**, **c**, **e–g**, **j–l**, data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistical significances ($P < 0.05$) were determined using one-way (**k**, **l**) and two-way (**b**, **c**, **e–g**, **j**) ANOVA followed by Tukey's multiple comparisons test, and was indicated by different letters. Source data are provided as a Source data file.

based transient assays in tobacco (Supplementary Fig. 3c), confirming that promoter polymorphisms drive differential induction of *LRP1* in response to bacterial and fungal pathogens.

To evaluate the functional consequences of CDS variation, we overexpressed *LRP1^{Tib}* and *LRP1^{Sic}* in Col under the constitutive 35S promoter. Transgenic lines (*LRP1^{Tib}OE-1/2*, *LRP1^{Sic}OE-1/2*) with comparable transgene expression levels (Supplementary Fig. 3d) exhibited distinct resistance phenotypes. *LRP1^{Tib}OE* lines displayed enhanced resistance to *Pst* DC3000, with reduced disease areas (Fig. 2b), lower bacterial titers (Fig. 2c, d), and elevated expression of the SA pathway marker *PR1* (Fig. 2e). In contrast, *LRP1^{Sic}OE* lines showed no improvement in *Pst* DC3000 resistance, but exhibited reduced susceptibility to *B. cinerea*, with smaller necrotic lesions (Fig. 2f, g), diminished fungal biomass (Fig. 2h), and increased expression of the camalexin biosynthesis gene *CYP71A13* (Fig. 2i), whereas *LRP1^{Tib}OE* lines showed susceptibility comparable to Col. These results indicate that amino acid substitutions in *LRP1* contribute to functional specificity: the Tib allele potentiates SA-mediated defense against bacteria, while the Sic allele enhances camalexin-dependent resistance to fungi.

To assess the contributions of promoter and coding variation, we introduced *pLRP1^{Tib}:LRP1^{Tib}*, *pLRP1^{Sic}:LRP1^{Sic}*, and their reciprocal promoter–coding swap constructs (*pLRP1^{Tib}:LRP1^{Sic}* and *pLRP1^{Sic}:LRP1^{Tib}*) into *lrp1^{Col}-1/2* mutants. *LRP1* induction followed promoter origin: the Tib promoter drove expression upon *Pst* DC3000 infection, whereas the Sic promoter was preferentially induced by *B. cinerea* (Supplementary Fig. 3e). In contrast, resistance phenotypes were specifically observed in the native promoter–coding combinations: Upon *Pst* DC3000 infection, *pLRP1^{Tib}:LRP1^{Tib} lrp1^{Col}* lines showed reduced bacterial growth and the highest *PR1* expression, whereas both swap lines were comparable to susceptible genotypes (Supplementary Fig. 3f, g). Following *B. cinerea* infection, *pLRP1^{Sic}:LRP1^{Sic} lrp1^{Col}* lines developed smaller lesions and showed the strongest *CYP71A13* induction, while swap lines showed increased susceptibility (Supplementary Fig. 3h–j). These results indicate that promoter variation controls pathogen-responsive *LRP1* expression, whereas effective defense responses depend on the matched promoter–coding combination.

LRP1 mediates immune signaling in response to *B. cinerea* and its elicitor (BCE)

Given the pivotal role of the coding region in shaping pathogen-specific immune responses and the limited understanding of the molecular function of *LRP1*, we first investigated the functional characteristics of *LRP1^{Col}*, which shares a similar coding region with *LRP1^{Sic}*, within immune signaling pathways. To this end, we generated *LRP1^{Col}* overexpression lines (*LRP1^{Col}OE-1/2*), T-DNA insertion mutants (*lrp1^{Col}-1/2*) that disrupt *LRP1^{Col}* function, and complementary lines in two *lrp1* mutant backgrounds (*pLRP1^{Col}:LRP1^{Col} lrp1^{Col}-1/2*) for pathogen inoculation assays (Supplementary Fig. 4a–c). Upon inoculation with *Pst* DC3000, *LRP1^{Col}* expression was not induced (Supplementary Fig. 4d), and no significant differences in disease susceptibility or

immune signaling were observed among these lines, suggesting that *LRP1^{Col}* is not involved in defense against *Pst* DC3000 (Supplementary Fig. 4f–i). However, following inoculation with *B. cinerea*, *LRP1^{Col}* expression was markedly upregulated (Supplementary Fig. 4e). The *LRP1^{Col}OE-1/2* lines displayed markedly enhanced resistance to *B. cinerea* compared to Col, while the *lrp1^{Col}-1/2* lines exhibited heightened susceptibility. Notably, the *pLRP1^{Col}:LRP1^{Col} lrp1^{Col}-1/2* lines restored resistance to levels comparable to those observed in Col (Fig. 3a, b). Increased expression of *CYP71A13* further supported these observations (Fig. 3c). The enhanced resistance phenotype observed in *LRP1^{Col}OE-1/2* lines, similar to that of *LRP1^{Sic}OE-1/2* lines, is likely attributable to the conserved coding regions shared by the two *LRP1* variants (Supplementary Fig. 4k). These results indicate that *LRP1^{Col}*, like *LRP1^{Sic}*, plays a crucial role in modulating immune responses to *B. cinerea*, highlighting its importance in pathogen-specific immunity, particularly in fungal pathogen defense.

To further investigate the molecular mode of action of *LRP1^{Col}*, we examined its subcellular localization. *LRP1^{Col}*, predominantly localized to the plasma membrane and endomembrane, consistent with PIP2a–mCherry, a marker of these compartments⁴¹, supporting its plasma membrane association despite the absence of a predicted transmembrane domain (Supplementary Fig. 5a). To test whether *LRP1^{Col}* functions in canonical bacterial PTI or elicitor-triggered immunity, we employed the D36E strain, which is a derivative of *Pst* DC3000 lacking all 36 effector genes¹⁰, alongside crude *B. cinerea* elicitor (BCE) to trigger immune responses. Immune responses to D36E did not significantly differ among the Col, *LRP1^{Col}OE-1/2*, *lrp1^{Col}-1/2*, and *pLRP1^{Col}:LRP1^{Col} lrp1^{Col}-1/2* lines. This was evidenced by comparable colony counts and consistent expression levels of *PR1*, along with key PTI markers, such as *WRKY22*, *WRKY29*, and *RBOHD* mRNA (Supplementary Fig. 5b, d). In contrast, BCE treatments elicited marked differences. The *LRP1^{Col}OE-1/2* lines exhibited significantly enhanced expressions of *CYP71A13*, *WRKY22*, *WRKY29*, and *RBOHD*, which were markedly diminished in *lrp1^{Col}-1/2* mutants but restored through complementation with *pLRP1^{Col}:LRP1^{Col}* (Fig. 3d, e). Additionally, BCE treatment induced robust production of ROS in *LRP1^{Col}OE-1/2* lines, whereas ROS production was significantly attenuated in *lrp1^{Col}-1/2* mutants compared to Col and *pLRP1^{Col}:LRP1^{Col} lrp1^{Col}-1/2* lines, both of which exhibited similarly strong ROS production (Fig. 3f). Western blot analysis further revealed increased phosphorylation of MPKs in *LRP1^{Col}OE-1/2* lines, confirming the role of *LRP1^{Col}* in mediating BCE-induced immune responses (Fig. 3g). These results underscore the specific involvement of *LRP1^{Col}* in BCE-triggered PTI responses, rather than in canonical bacterial PTI, and highlight its critical role in modulating pathogen-specific immune pathways.

Given that BAK1 and SOBIR1 act as central signaling components in many LRR-mediated cell-surface immune pathways, we next examined whether they are involved in *LRP1^{Col}*-mediated immune signaling. BCE-induced ROS production was assessed in *LRP1^{Col}OE bak1-5 bkl1* and *LRP1^{Col}OE sobir1-12* plants. Compared to Col and *LRP1^{Col}* plants, BCE-induced ROS production was significantly reduced in both

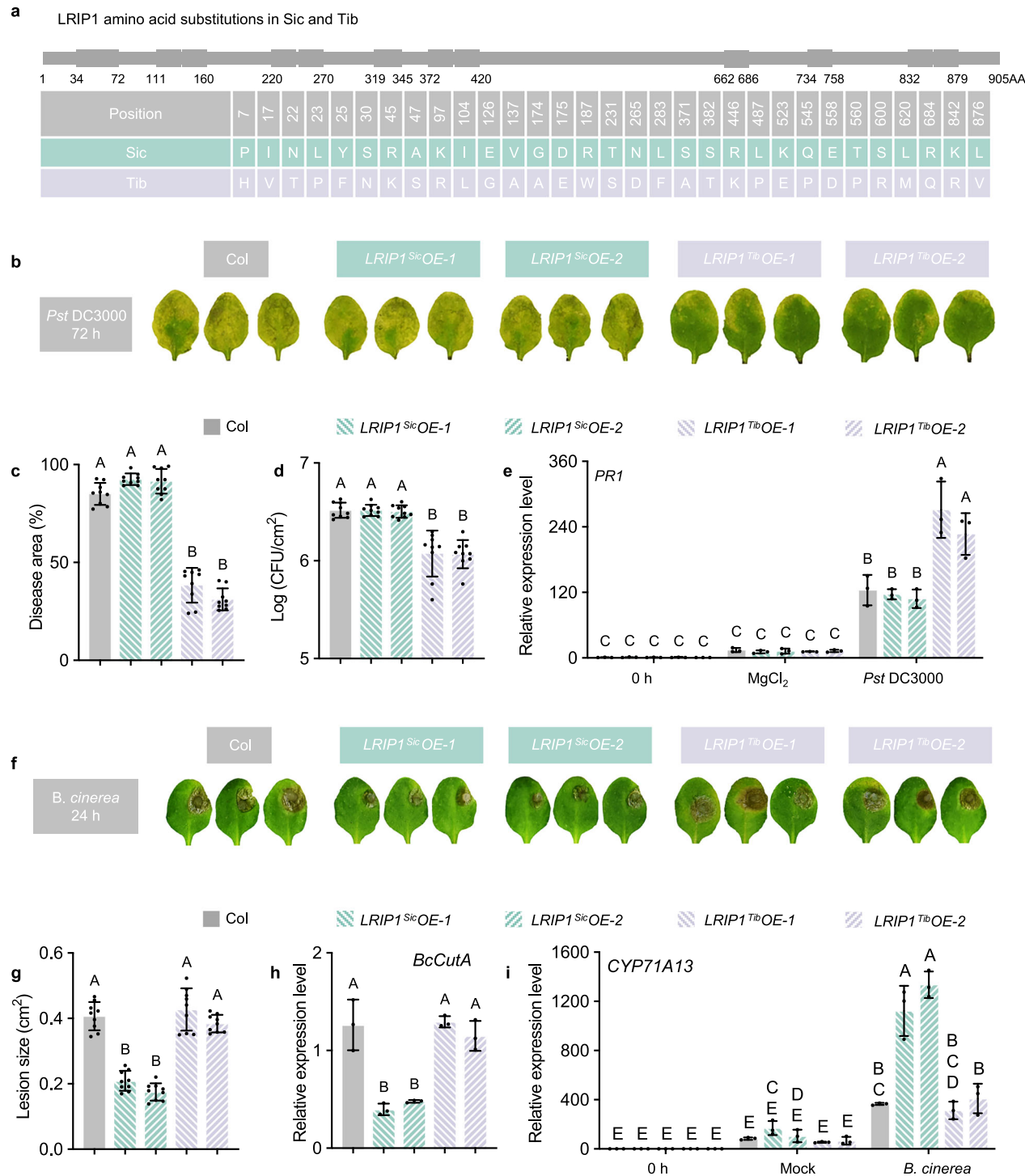


Fig. 2 | LRIP1 mediates defense responses through pathogen-induced expression and signaling. **a** Schematic representation of the LRIP1 coding region, highlighting allelic variations between Sic and Tib accessions. The line represents the coding region, while the rectangle indicates the LRR domain. The numbers indicate the position of the domain, and AA represents amino acids. **b** Disease area development in Col, LRIP1^{Sic}OE-1/2 and LRIP1^{Tib}OE-1/2 lines at 72 h post-inoculation with *Pst* DC3000. Quantification of disease area percentages relative to total leaf area in (b, c) and bacterial CFUs at 48 h post-inoculation with *Pst* DC3000 (d) in the indicated lines. **e** Expression levels of *PR1* in the indicated lines before and 24 h after inoculation with *Pst* DC3000 or MgCl₂ (control). **f** Lesion development in Col,

LRIP1^{Sic}OE-1/2 and LRIP1^{Tib}OE-1/2 lines at 24 h post-inoculation with *B. cinerea*. **g** Quantification of lesion size in (f). **h** Expression of *BcCutA* in the indicated lines at 24 h following inoculation with *B. cinerea*. **i** Expression levels of *CYP71A13* in the indicated lines at 24 h post-inoculation with *B. cinerea* or mock controls (equal-sized MS agar blocks). In c–e, g–i, data are presented as means ± s.d. (*n* = 3 biological replicates). Statistically significant differences (*P* < 0.05) were determined using one-way (c, d, g, h) and two-way (e, i) ANOVA with Tukey’s multiple comparisons test and are indicated by different letters. All transgenic lines were generated in the Col background. Source data are provided as a Source data file.

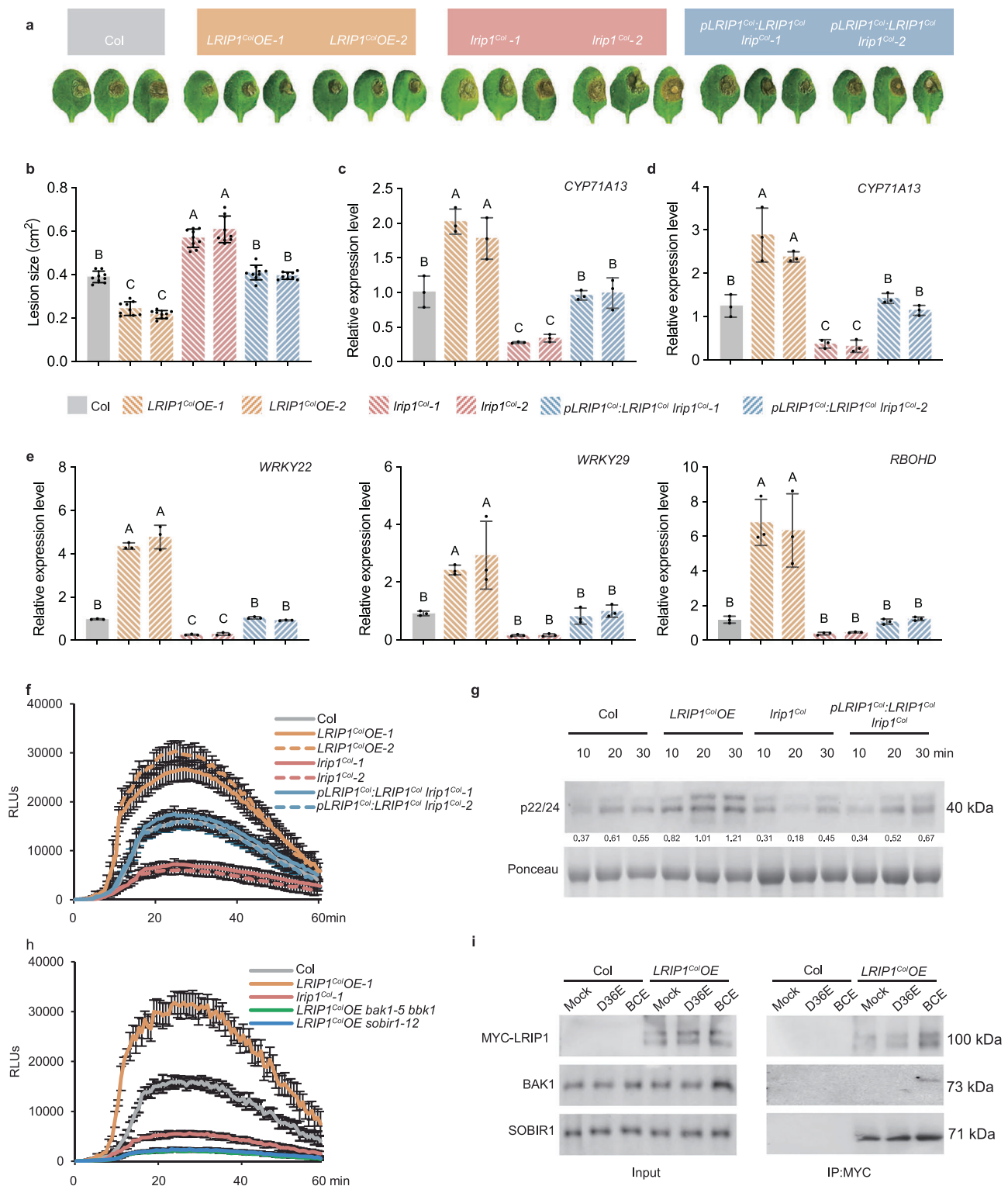


Fig. 3 | *LIRIP1* in Col is responsible to resist *B. cinerea*. **a** Lesion development in Col, *LRIP1OE-1/2*, *lrip1-1/2* (mutants) and complemented lines (*pLRIP1::LRIP1 lrip1-1/2*) at 20 h post-inoculation with *B. cinerea*. **b** Quantification of the lesion size shown in (a). Expression levels of *CYP71A13* in the indicated lines at 24 h post-inoculation with *B. cinerea* (c) or BCE (d). **e** Expression levels of *WRKY22*, *WRKY29*, and *RBOHD* in the indicated lines at 24 h post-inoculation with BCE. **f** ROS production detected in the indicated lines after BCE treatment. **g** Protein levels of phosphorylated MPKs in the indicated lines after BCE treatment at the indicated time points. Phosphorylated MPK6 (top band), MPK3 (middle band), and MPK4/MPK11 (bottom band) in *Arabidopsis* were detected by immunoblotting using an anti-p44/42 MAPK antibody. Ponceau staining was used as a loading control. **h** ROS production in Col, *LRIP1OE-1*, *lrip1Col-1*, and complemented lines (*pLRIP1::LRIP1Col-1* and *pLRIP1::LRIP1Col-2*) following BCE treatment. **i** Co-IP analysis of *LIRIP1* interactions with BAK1 and SOBIR1. Total proteins were isolated from *LRIP1OE* plants (MYC-tagged *LIRIP1*) before and after D36E or BCE treatments at 30 min. Immunoprecipitation was performed using anti-MYC beads, and immunoblotting was conducted with anti-MYC, anti-BAK1, and anti-SOBIR1 antibodies. In **g**, **i**, proteins were separated on a 12% SDS-PAGE gel. Two independent experiments produced similar results. In **b–e**, data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistically significant differences ($P < 0.05$) were determined using one-way ANOVA with Tukey's multiple comparisons test and are indicated by different letters. Source data are provided as a Source data file.

LRIP1OE-1, *lrip1-1*, *LRIP1OE-1 bak1-5 bkk1*, *LRIP1OE-1 sobir1-12* lines following BCE treatment. **i** Co-IP analysis of *LIRIP1* interactions with BAK1 and SOBIR1. Total proteins were isolated from *LRIP1OE* plants (MYC-tagged *LIRIP1*) before and after D36E or BCE treatments at 30 min. Immunoprecipitation was performed using anti-MYC beads, and immunoblotting was conducted with anti-MYC, anti-BAK1, and anti-SOBIR1 antibodies. In **g**, **i**, proteins were separated on a 12% SDS-PAGE gel. Two independent experiments produced similar results. In **b–e**, data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistically significant differences ($P < 0.05$) were determined using one-way ANOVA with Tukey's multiple comparisons test and are indicated by different letters. Source data are provided as a Source data file.

LRIP1^{Col}OE bak1-5 bkk1 and *LRIP1^{Col}OE sobir1-12* lines, indicating that *BAK1* and *SOBIR1* are essential components of *LRIP1^{Col}*-dependent immune signaling (Fig. 3h). Co-immunoprecipitation (Co-IP) assays confirmed that *LRIP1^{Col}* interacts with *BAK1* and *SOBIR1*. *LRIP1^{Col}* forms a constitutive complex with *SOBIR1* under all conditions, whereas its interaction with *BAK1* is specifically induced following BCE treatment (Fig. 3i). These findings suggest that *LRIP1^{Col}* functions through co-receptor-associated complexes.

Functional divergence of two *LRIP1* alleles in mediating immune responses to different pathogens

Having established that *LRIP1^{Col}* is involved in immune responses, we further investigated the distinct roles of *LRIP1^{Sic}* and *LRIP1^{Tib}* in mediating immunity against different pathogens. Consistent with *LRIP1^{Col}*, both *LRIP1^{Sic}* and *LRIP1^{Tib}* were predominantly associated with the plasma membrane (Supplementary Fig. 5a). To further investigate the function of *LRIP1^{Sic}*, we generated *lrp1^{Sic}* mutants using the CRISPR-Cas9 system and evaluated their responses to *Pst* DC3000, D36E, *B. cinerea*, and BCE (Supplementary Fig. 6a). Inoculation with *Pst* DC3000 and D36E revealed no significant differences in disease susceptibility or *PR1* mRNA levels among these lines, indicating that *LRIP1^{Sic}* does not contribute to immunity against *Pst* DC3000 (Supplementary Fig. 6b, c). In contrast, following *B. cinerea* inoculation, *lrp1^{Sic}-1/2* mutants exhibited significantly larger lesion areas, while *LRIP1^{Sic}OE-1/2* lines displayed markedly smaller lesion areas. Complementation of *lrp1^{Sic}-1/2* mutants with *pLRIP1^{Sic}:LRIP1^{Sic}* effectively restored the wild-type phenotype, underscoring the critical role of *LRIP1^{Sic}* in immunity against *B. cinerea* (Fig. 4a, b). Reduced *CYP71A13* induction in *lrp1^{Sic}-1/2*, alongside increased *CYP71A13* expression following *B. cinerea* infection in *LRIP1^{Sic}OE-1/2*, further emphasized *LRIP1^{Sic}*'s specific role in mediating defense responses against this pathogen (Fig. 4c). BCE treatment elicited robust immune responses, with *LRIP1^{Sic}OE-1/2* lines showing significantly elevated expression of *CYP71A13*, *WRKY22*, *WRKY29* and *RBOHD* (Fig. 4d and Supplementary Fig. 6d). These responses were substantially diminished in *lrp1^{Sic}-1/2* mutants but were successfully restored upon complementation with *pLRIP1^{Sic}:LRIP1^{Sic}* (Fig. 4d and Supplementary Fig. 6d). Additionally, BCE treatment induced strong ROS production in *LRIP1^{Sic}OE-1/2* lines, whereas ROS levels were markedly reduced in *lrp1^{Sic}-1/2* mutants (Fig. 4e). These findings identify *LRIP1^{Sic}* as a key regulator of BCE-induced immune signaling. To assess the specificity of *LRIP1^{Sic}*-mediated responses, we tested whether *LRIP1^{Sic}* contributes to chitin-triggered signaling. No detectable differences in *WRKY29* induction were observed following chitin treatment (Supplementary Fig. 6e).

Next, we assessed the role of *LRIP1^{Tib}* in responses to *Pst* DC3000, D36E, *B. cinerea*, and BCE. Overexpression of *LRIP1^{Tib}* in its native Tib background further enhanced immune responses against both *Pst* DC3000 and D36E (Fig. 4f–j and Supplementary Fig. 7a). *LRIP1^{Tib}OE-1/2* lines displayed increased expression of *PR1*, *WRKY22*, *WRKY29*, and *RBOHD* (Fig. 4g, i, and Supplementary Fig. 7b), along with significantly elevated ROS production compared to the native Tib background (Fig. 4j). These results support *LRIP1^{Tib}* as a positive regulator of immune responses, particularly in D36E-triggered immune responses. The repeated failure in generating effective *lrp1^{Tib}* mutants after six transformation attempts with the CRISPR-Cas9 system necessitated an alternative approach. Complementary lines expressing *pLRIP1^{Tib}:LRIP1^{Tib}* were developed in Col and Sic backgrounds, as both Col and Sic lacked functional *LRIP1* in DC3000-triggered immune signaling, effectively serving as native *lrp1^{Tib}* mutants. These *pLRIP1^{Tib}:LRIP1^{Tib}-1/2* lines in Sic and Col backgrounds demonstrated markedly enhanced resistance to both *Pst* DC3000 and D36E. Resistance to *Pst* DC3000 was characterized by reduced disease areas, diminished bacterial colony numbers, and elevated *PR1* mRNA expression, highlighting the critical role of *LRIP1^{Tib}* in DC3000-triggered immunity (Figs. 1k and 4k, l). Similarly, enhanced

resistance to D36E was evidenced by significantly reduced bacterial loads and increased expression of key defense related genes, including *PR1*, *WRKY22*, *WRKY29* and *RBOHD* (Fig. 4m, n, and Supplementary Fig. 7c). Robust ROS production further validated heightened immune activation in these lines, underscoring the integral role of *LRIP1^{Tib}* in D36E-triggered immune responses (Fig. 4o). Inoculation with *B. cinerea* and BCE showed no notable differences in *CYP71A13* mRNA levels across these lines, suggesting that *LRIP1^{Tib}* does not play a role in *B. cinerea* immunity (Supplementary Fig. 7d, e). To assess whether *LRIP1^{Tib}*-mediated responses involve canonical bacterial PAMP signaling, we examined *WRKY29* expression following treatment with flg22, elf18, and nlp20. None of these elicitors resulted in detectable *LRIP1^{Tib}*-dependent differences in *WRKY29* induction (Supplementary Fig. 7f).

Amino acid variation in *LRIP1* modulates elicitor-dependent immune signaling

To further explore the role of amino acid variations in *LRIP1* in determining signaling specificity, we examined how these mutations influence immune responses in *LRIP1^{Sic}OE-1/2* and *LRIP1^{Tib}OE-1/2* lines, using the Col background to minimize background effects and ensure consistent and reliable comparisons. Given that *LRIP1* associates with *SOBIR1* and *BAK1*, we next investigated whether amino acid variations in *LRIP1* affect the specificity of co-receptor recruitment in response to different elicitors. Co-IP assays performed before and after treatment with D36E or BCE revealed no significant differences in the interactions of *LRIP1^{Sic}* and *LRIP1^{Tib}* with *SOBIR1* (Fig. 5a). However, the interaction of *LRIP1^{Sic}* with *BAK1* was specifically induced by BCE treatment, while *LRIP1^{Tib}* interacted with *BAK1* only following D36E treatment (Fig. 5a). Similarly, split-luciferase complementation assays confirmed that *LRIP1^{Sic}* forms a constitutive complex with *SOBIR1*, while its interaction with *BAK1* was specifically induced by BCE treatment. In contrast, *LRIP1^{Tib}* also consistently interacted with *SOBIR1* but formed a complex with *BAK1* only upon D36E treatment (Fig. 5b). These findings suggest that amino acid variations in *LRIP1* modulate elicitor-dependent recruitment of *BAK1*, leading to qualitatively different downstream immune programs.

Given these observations, we hypothesized that the *LRIP1* allelic variants might differ in their ability to respond to distinct pathogen-derived elicitors. Supporting this hypothesis, inoculation with D36E revealed that *LRIP1^{Tib}OE-1/2* lines exhibited significantly reduced bacterial colony counts compared to *LRIP1^{Sic}OE-1/2* and Col, which displayed similar colony numbers (Fig. 5c). Furthermore, *LRIP1^{Tib}OE-1/2* lines showed elevated ROS production (Fig. 5d), enhanced phosphorylation of MPKs (Fig. 5e), and increased mRNA levels of *PR1*, *WRKY22*, *WRKY29*, and *RBOHD* (Fig. 5f) compared to *LRIP1^{Sic}OE-1/2* and Col, which exhibited comparable levels of these immune response markers. These results indicate that *LRIP1^{Tib}* preferentially mediates immune responses associated with D36E treatment. Conversely, inoculation with BCE demonstrated that *LRIP1^{Sic}OE-1/2* lines displayed significantly elevated ROS production (Fig. 5g), enhanced phosphorylation of MPKs (Fig. 5h), and increased expression of *CYP71A13*, *WRKY22*, *WRKY29*, and *RBOHD* (Fig. 5i) compared to *LRIP1^{Tib}OE-1/2* and Col, which exhibited comparable levels of these responses. These findings suggest that *LRIP1^{Sic}* preferentially mediates immune responses triggered by BCE treatment. Moreover, we utilized AlphaFold3 to predict the structures of *LRIP1^{Sic}-BAK1-SOBIR1* and *LRIP1^{Tib}-BAK1-SOBIR1* complexes. Structural analyses revealed distinct differences between the two complexes, which may be associated with their differential ability to recruit co-receptors (Supplementary Fig. 8a).

To further define how elicitor-specific differences in co-receptor recruitment are translated into downstream signaling outputs, we performed RNA-seq analysis following 3 h BCE or D36E treatments (Supplementary Fig. 8b, c, and Supplementary Data 5). Following D36E treatment, *LRIP1^{Tib}* showed preferential transcriptional responses,

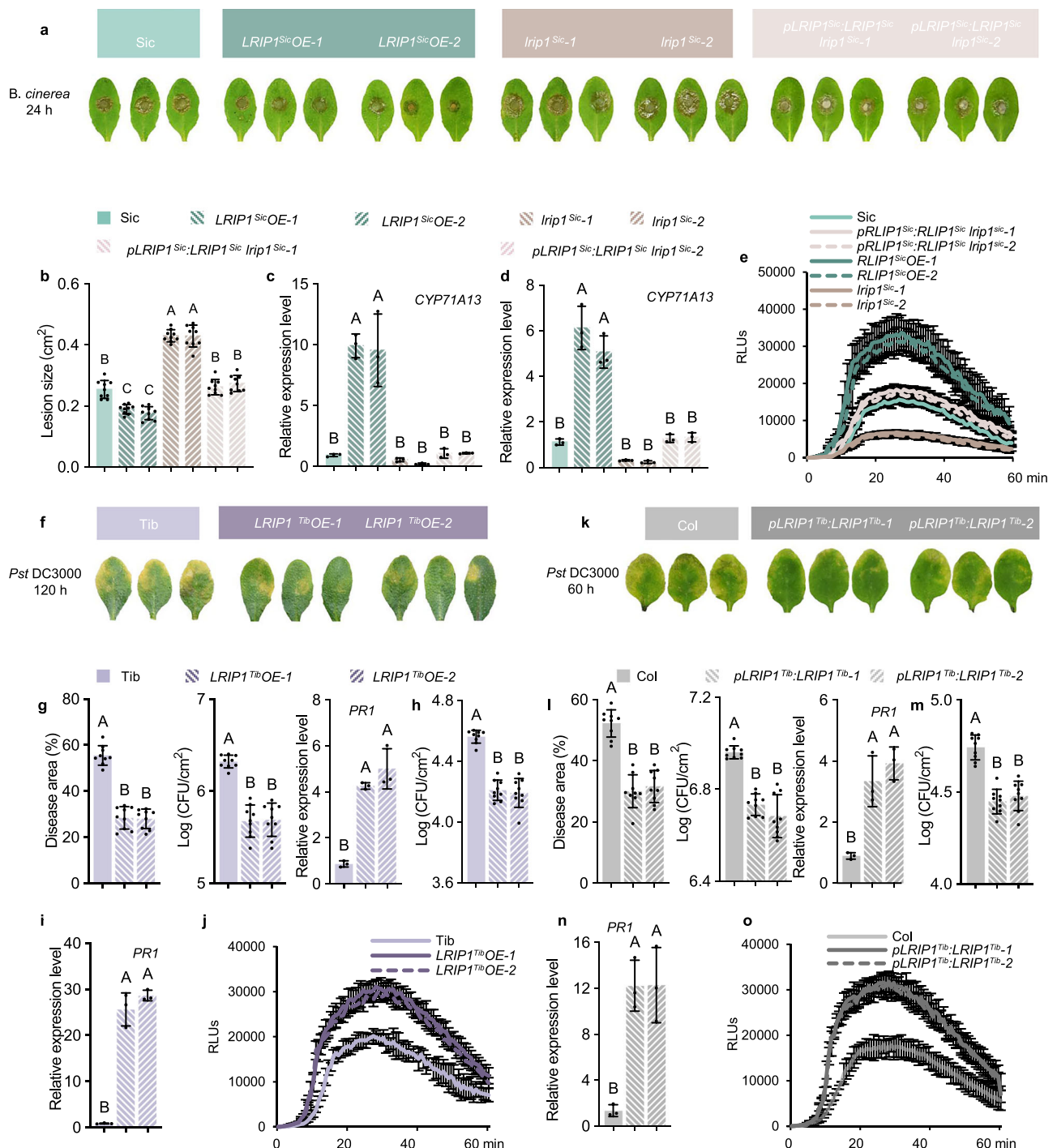
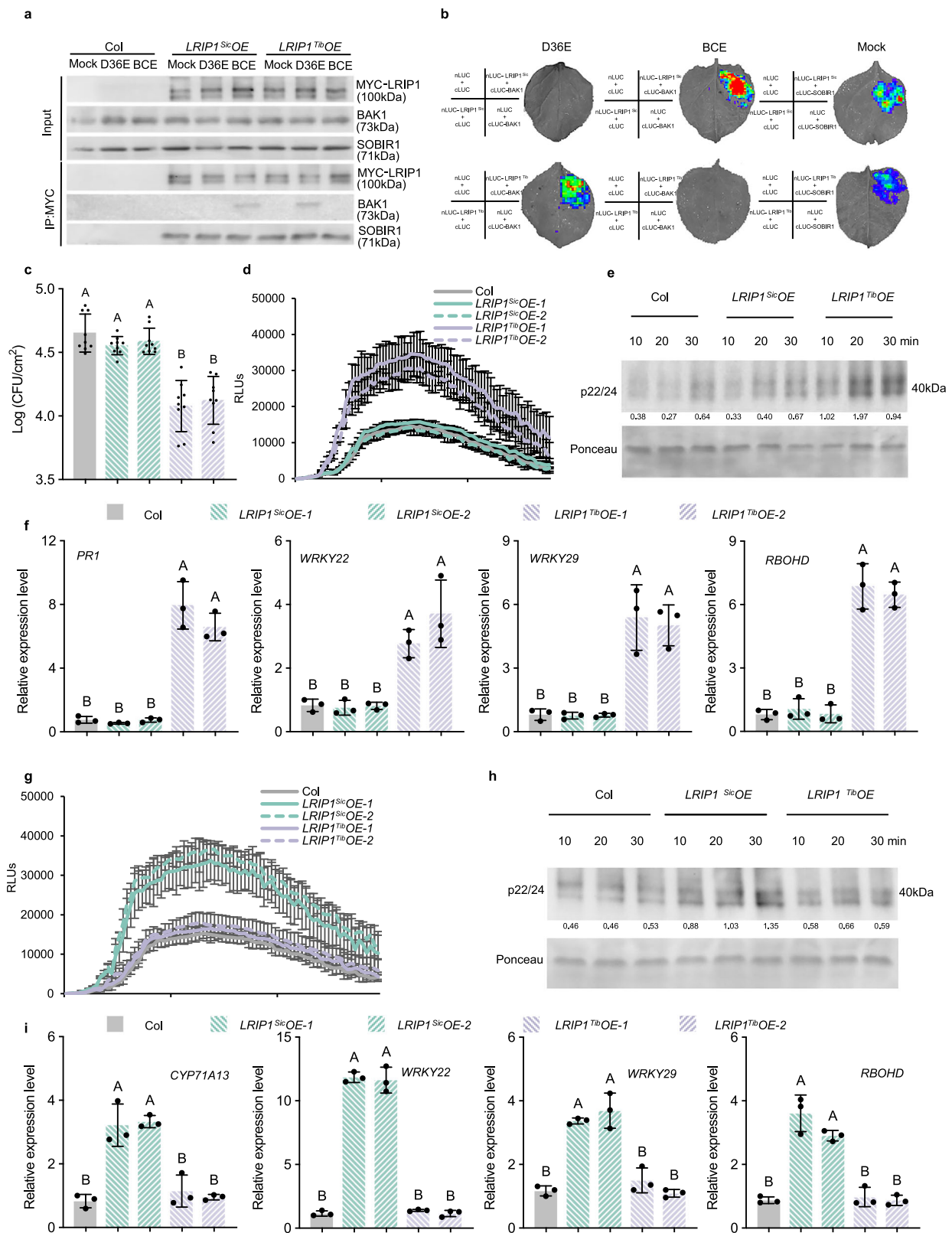


Fig. 4 | $LRIP1^{Sic}$ and $LRIP1^{Tib}$ function in diverse pathogens resistance in Sic and Tib backgrounds. **a** Lesion development in Sic, $LRIP1^{Sic}OE-1/2$, $lrip1^{Sic-1/2}$ mutants and complemented lines ($pLRIP1^{Sic}:LRIP1^{Sic}$ $lrip1^{Sic-1/2}$) at 24 h post-inoculation with *B. cinerea*. All the transgenic lines were generated in the Sic background.

b Quantification of the lesion size shown in (a). Expression levels of *CYP71A13* in the indicated lines at 24 h post-inoculation with *B. cinerea* (c) or BCE (d). **e** ROS production in the indicated lines following BCE treatment. **f** Disease area development in Tib, $LRIP1^{Tib}OE-1/2$ lines inoculated with *Pst* DC3000 at 96 h post-inoculation. $LRIP1^{Tib}OE-1/2$ lines were generated in the Tib background. **g** Quantification of disease area percentages relative to total leaf area (left), bacterial CFUs (middle) at 48 h post-inoculation, and *PR1* expression levels (right) at 24 h post-inoculation with *Pst* DC3000 in Tib and $LRIP1^{Tib}OE-1/2$ lines. Bacterial CFUs at 48 h post-inoculation (h) and *PR1* expression levels at 24 h post-inoculation (i) with D36E in

Tib and $LRIP1^{Tib}OE-1/2$ lines. **j** ROS production in Tib and $LRIP1^{Tib}OE-1/2$ lines after D36E treatment. **k** Disease area development in Col and complemented $pLRIP1^{Tib}:LRIP1^{Tib-1/2}$ in the Col background 60 h post-inoculation with *Pst* DC3000. **l** Quantification of disease area percentages relative to total leaf area (left), bacterial CFUs (middle) at 48 h post-inoculation, and *PR1* expression levels (right) at 24 h post-inoculation with *Pst* DC3000 Col and $pLRIP1^{Tib}:LRIP1^{Tib-1/2}$ lines. Bacterial CFUs at 48 h post-inoculation (m) and *PR1* expression levels at 24 h post-inoculation (n) with D36E in Col and $pLRIP1^{Tib}:LRIP1^{Tib-1/2}$ lines. **o** ROS production in Col and $pLRIP1^{Tib}:LRIP1^{Tib-1/2}$ lines after D36E treatment. In b–d, g–i, l–n, data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistically significant differences ($P < 0.05$) were determined using one-way ANOVA with Tukey's multiple comparisons test and are indicated by different letters. Source data are provided as a Source data file.



characterized by the induction of receptor-like genes (e.g., *PEPR1*, *RLPs*, and *LRR-RLKs*), calcium- and MAPK-related signaling components (*CNGC3*, *MPK3*, and *MAPKKK19*), and transcription factors associated with immune regulatory pathways, including the SA-associated gene *NIMIN1*.

In contrast, following BCE treatment, *LRIP1^{Sic}* preferentially induced genes associated with early immune signaling. Several

receptor-like proteins and kinases associated with PTI (including *RLPs*, *WAK-like proteins*, and *LRR-RLKs*) were transcriptionally upregulated specifically in *LRIP1^{Sic}*. Consistent with this, *LRIP1^{Sic}* showed increased expression of downstream MAPK components (*MKK7*, *MAPKKK15*), defense-associated transcription factors (*WRKY33*, *MYB51*), and genes involved in defense-related metabolism. Together, these results indicate that *LRIP1^{Sic}* and *LRIP1^{Tb}* undergo early transcriptional divergence

Fig. 5 | *LRIP1^{Sic}OE* and *LRIP1^{Tib}OE* lines in response to BCE- and D36E-triggered immunity, respectively. **a** Co-IP of *LRIP1^{Sic}* and *LRIP1^{Tib}* interactions with BAK1 and SOBIR1. Total proteins were extracted from Col, *LRIP1^{Sic}OE*, and *LRIP1^{Tib}OE* lines treated with or without D36E or BCE for 30 min. MYC-tagged *LRIP1* proteins were immunoprecipitated using anti-MYC agarose beads, and immunoblotting was performed with anti-MYC, anti-BAK1, and anti-SOBIR1 antibodies. Input controls show total protein levels for MYC-*LRIP1*, BAK1, and SOBIR1. Col was used as a negative control. **b** Split-luciferase complementation assay showing interactions between *LRIP1^{Sic}* and *LRIP1^{Tib}* with BAK1 and SOBIR1. Leaves were examined 30 min after infiltration with D36E (left) or BCE (middle) to detect interactions of *LRIP1^{Sic}*-BAK1 and *LRIP1^{Tib}*-BAK1, respectively. Untreated leaves (right) were analyzed to assess interactions of *LRIP1^{Sic}*-SOBIR1 and *LRIP1^{Tib}*-SOBIR1. Luminescence signals indicate protein–protein interactions. **c** Bacterial CFUs in Col, *LRIP1^{Sic}OE-1/2* and *LRIP1^{Tib}OE-1/2* lines at 48 h post-inoculation with D36E. **d** ROS production in Col, *LRIP1^{Sic}OE-1/2* and *LRIP1^{Tib}OE-1/2* lines following D36E treatment. **e** MAPK

phosphorylation in Col, *LRIP1^{Sic}OE*, and *LRIP1^{Tib}OE* lines at different time points after D36E inoculation. Phosphorylation of MPKs was detected using anti-p-p44/42 antibody. Equal amounts of total proteins were loaded into gel lanes. **f** Expression of *PR1*, *WRKY22*, *WRKY29*, and *RBOHD* in Col, *LRIP1^{Sic}OE-1/2*, and *LRIP1^{Tib}OE-1/2* at 24 h post-inoculation with D36E. **g** ROS production in Col, *LRIP1^{Sic}OE-1/2*, and *LRIP1^{Tib}OE-1/2* lines following BCE treatment. **h** MAPK phosphorylation was detected using anti-p-p44/42 antibody, in Col, *LRIP1^{Sic}OE* and *LRIP1^{Tib}OE* lines at different time points after BCE inoculation. Equal amounts of total proteins were loaded into gel lanes. **i** Expression of *CYP71A13*, *WRKY22*, *WRKY29*, and *RBOHD* in Col, *LRIP1^{Sic}OE-1/2*, and *LRIP1^{Tib}OE-1/2* lines at 24 h after BCE inoculation. In **a**, **e**, **h**, proteins were separated on 12% (**a**) or 15% (**e**, **h**) SDS-PAGE gels. Two independent experiments yielded similar results. In **c**, **f**, **i**, data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistical significance was determined by one-way ANOVA followed by Tukey's multiple-comparisons test ($P < 0.05$). Source data are provided as a Source data file.

in response to distinct elicitor contexts, leading to qualitatively different downstream immune programs.

The SNPs in the LRR domain affect the function of *LRIP1* proteins, thereby influencing elicitor-specific immune signaling

To further investigate the functional impact of specific coding region variations on *LRIP1*, we constructed segment-replacement vectors and generated eight point-mutation vectors targeting distinct variation sites within the LRR domain (Fig. 6a). Stable *A. thaliana* lines with comparable *LRIP1* expression levels as *LRIP1^{Sic}OE-1* and *LRIP1^{Tib}OE-1* lines were established to ensure reliable results (Supplementary Fig. 9a).

Inoculation with *Pst* DC3000 and D36E revealed intermediate bacterial colony counts in *LRIP1^{Sic}OE-1/2* and *LRIP1^{Tib}OE-1/2* lines. The colony counts in these lines fell between the lowest levels observed in *LRIP1^{Tib}OE-1* line and the highest levels observed in Col, *LRIP1^{Sic}OE-1*, and *LRIP1^{Tib}OE-1/2* lines, which exhibited comparable colony counts for both pathogens (Fig. 6b and Supplementary Fig. 9b). Consistent trends were observed in the analysis of *PR1* expression levels, further supporting these findings (Fig. 6c and Supplementary Fig. 9c). Notably, *LRIP1^{Sic}8mOE-1/2* lines exhibited bacterial colony counts and *PR1* expression levels comparable to *LRIP1^{Tib}OE-1* line, while *LRIP1^{Tib}8mOE-1/2* lines displayed values comparable to Col and *LRIP1^{Sic}OE-1* lines (Fig. 6d, e, and Supplementary Fig. 9d). Additionally, introducing the *proLRIP1^{Tib}:LRIP1^{Tib}8m* construct into the *Sic* background did not enhance resistance to *Pst* DC3000 or D36E, as evidenced by similar colony counts and *PR1* mRNA levels (Fig. 6f, g, and Supplementary Fig. 9e, f). These findings suggest that loci associated with Tib-mediated resistance to *Pst* DC3000 are located at K45, S47, G126, S231, D265, and T382. Since S231 is identical to Col, further analysis focused on the remaining five loci. Overexpression lines carrying single *Sic*-to-Tib point mutations at these positions were generated for further analysis (Supplementary Fig. 9g). *Pst* DC3000 or D36E inoculation experiments revealed that mutations at K45, S47, G126, and T382 significantly enhanced *LRIP1^{Sic}* resistance. However, the resistance conferred by these mutations did not reach the levels observed in *LRIP1^{Tib}*, suggesting that their combined effects are essential for Tib-mediated resistance (Fig. 6h, i, and Supplementary Fig. 9h, i).

Inoculation with *B. cinerea* demonstrated intermediate lesion sizes in *LRIP1^{Sic}OE-1/2* and *LRIP1^{Tib}OE-1/2* lines, which fell between the smallest lesions observed in *LRIP1^{Sic}OE-1* line and the largest lesions in Col, *LRIP1^{Tib}OE-1*, and *LRIP1^{Tib}OE-1/2* lines, all of which exhibited similar lesion areas (Supplementary Fig. 10a, b). These results were corroborated by *CYP71A13* expression levels following *B. cinerea* and BCE treatments (Supplementary Fig. 10c and Fig. 6j). *LRIP1^{Sic}OE-1* and *LRIP1^{Tib}8mOE-1/2* lines exhibited smaller lesion areas (after *B. cinerea* inoculation) and higher *CYP71A13* expression levels (after *B. cinerea* and BCE treatments) compared to Col, *LRIP1^{Sic}8mOE-1/2*, and *LRIP1^{Tib}OE-1* lines (Supplementary Fig. 10d, e, and Fig. 6k). Introducing

the *proLRIP1^{Sic}:LRIP1^{Sic}8m* construct into the Tib background did not improve resistance to *B. cinerea*, as indicated by similar lesion areas and *CYP71A13* mRNA levels (Supplementary Fig. 10f and Fig. 6l). These findings suggest that loci determining *Sic*-mediated resistance to *B. cinerea* are located at R684 and L876. Single Tib-to-*Sic* point mutations at these positions significantly enhanced resistance to *B. cinerea* (Fig. 6m). However, the resistance level remained below that of *LRIP1^{Sic}*, indicating that the combined effects of these mutations are critical for *Sic*-mediated resistance.

In summary, these results demonstrate that specific loci within the LRR domain of *LRIP1* play pivotal roles in modulating immune responses to diverse pathogens. These findings highlight the critical function of *LRIP1* in pathogen resistance.

Two distinct *LRIP1* haplotype groups correlate with local precipitation adaptation

To investigate patterns of natural variation at the *LRIP1* locus in *A. thaliana*, we analyzed whole-genome sequencing data from 737 geographically diverse accessions, focusing on variation within its functional domains. The whole genome sequencing data revealed exceptional nucleotide diversity ($\pi > 0.01$) in the *LRIP1* coding region, exceeding 98% of surveyed genes in the species. Phylogenetic analysis revealed two deeply diverged lineages of *LRIP1* (Supplementary Fig. 11a), which were further supported by haplotype network analysis, defining two major groups corresponding to the Tib and *Sic* haplotypes (Supplementary Fig. 11b). Furthermore, the Tib allele occupies a relatively basal position within *A. thaliana* (Supplementary Fig. 11c). Geographically, Tib-associated accessions are predominantly found in arid environments, whereas *Sic* haplotypes are more frequently associated with humid regions (Supplementary Fig. 11d). This bioclimatic segregation suggests differential selection pressures across moisture gradients, potentially driving *LRIP1* allelic diversification. To test functional consequences of this divergence, we performed further pathogen resistance assays on representative accessions under controlled conditions. All accessions of the Tib haplotype group conferred enhanced resistance to *Pst* DC3000, showing a reduction in bacterial growth compared to *Sic* carriers (Fig. 7a, b). Conversely, the accessions of the *Sic* haplotypes exhibited superior defense against *B. cinerea*, with smaller lesion areas than Tib counterparts (Fig. 7c). Significant variation in resistance to both pathogens was observed across ecotypes. This specialization may reflect adaptation to ecological niches where one pathogen type exerts dominant selective pressure, favoring optimization of specific resistance mechanisms over broad-spectrum immunity.

Discussion

Our study identifies *LRIP1* as a putative non-canonical LRR immune protein in *A. thaliana*, lacking both an intracellular kinase domain and a clearly predicted transmembrane region. Structural and functional

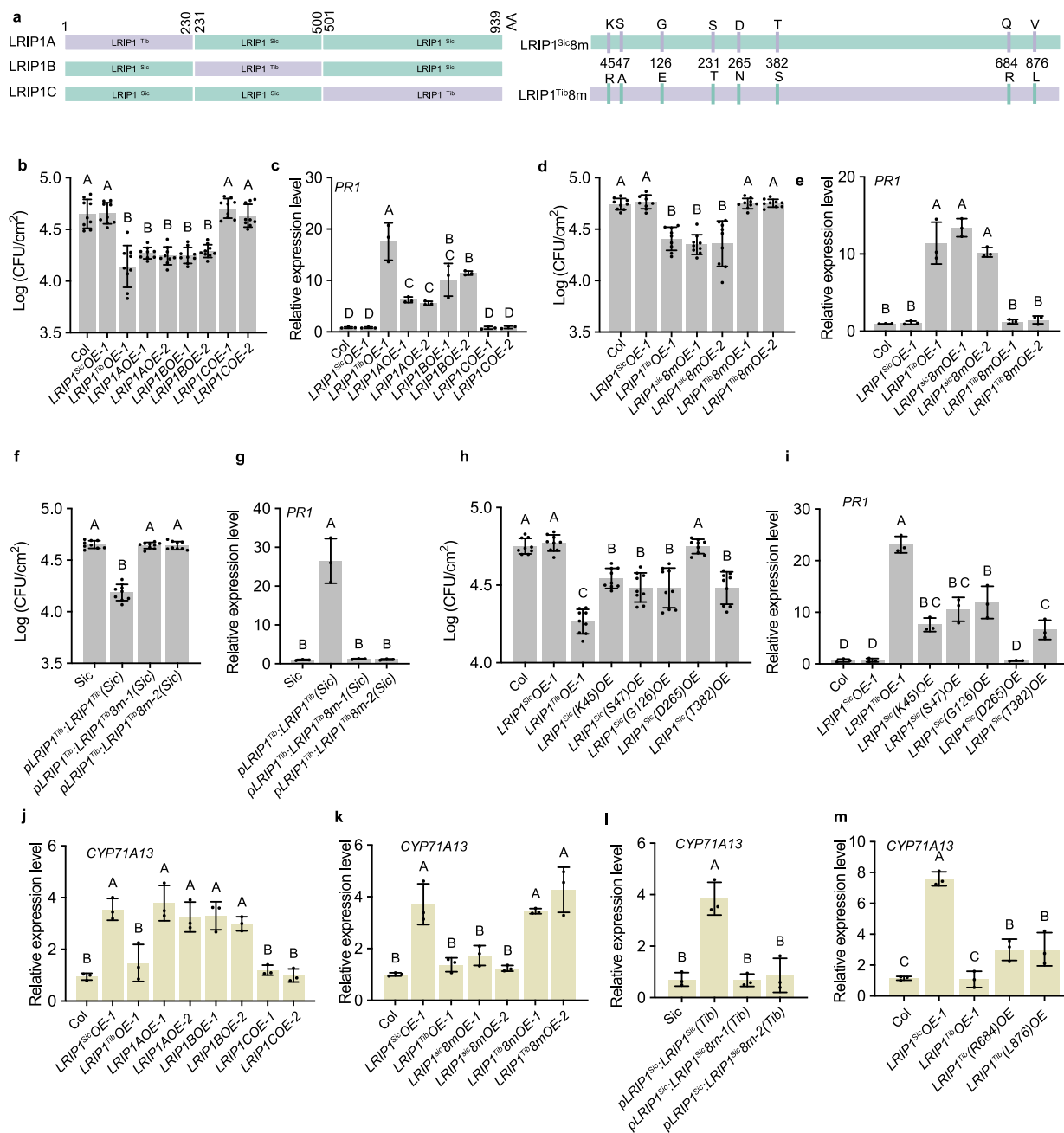


Fig. 6 | Specific variants in *LRIP1^{Sic}* and *LRIP1^{Tib}* confer immunity to D36E and BCE. **a Schematic representation of segment replacements in the coding regions of *LRIP1A*, *LRIP1B*, and *LRIP1C* (left) and amino acid substitutions in the LRR domain of *LRIP1^{Sic}8m* and *LRIP1^{Tib}8m* (right). Green and purple indicate *LRIP1^{Sic}* and *LRIP1^{Tib}* coding regions, respectively. Bacterial CFUs (**b**) and *PR1* expression (**c**) in *LRIP1^{Sic}OE-1/2*, *LRIP1^{Tib}OE-1/2*, and control lines at 48 and 24 h post-inoculation with D36E, respectively. Bacterial CFUs (**d**) and *PR1* expression (**e**) in *LRIP1^{Sic}8mOE-1/2*, *LRIP1^{Tib}8mOE-1/2*, and control lines after D36E inoculation. Bacterial CFUs (**f**) and *PR1* expression (**g**) in *Sic*, *pLRIP1^{Tib}:LRIP1^{Tib}(Sic)*, and *pLRIP1^{Tib}:LRIP1^{Tib}8m-1/2(Sic)* lines after D36E inoculation. *Sic* and *pLRIP1^{Tib}:LRIP1^{Tib}(Sic)* were used as controls. Bacterial CFUs (**h**) and *PR1* expression (**i**) in various single substitution lines after D36E inoculation. *LRIP1^{Sic}(K45/S47/G126/S231/D265/T382)* represents the substitution of Haplotype-Sic AAs with Hap-Tib AAs at specific positions in the *LRIP1^{Sic}* coding region. **j** *CYP71A13* expression in *LRIP1^{Sic}OE-1/2*, *LRIP1^{Tib}OE-1/2*, and control lines at 24 h post-inoculation**

with BCE. **k** *CYP71A13* expression in *LRIP1^{Sic}8mOE-1/2*, *LRIP1^{Tib}8mOE-1/2*, and control lines at 24 h post-inoculation with BCE. **l** *CYP71A13* expression in Tib, *pLRIP1^{Sic}:LRIP1^{Sic}(Tib)*, *pLRIP1^{Sic}:LRIP1^{Sic}8m(Tib)* lines at 24 h post-inoculation with BCE. Tib and *pLRIP1^{Sic}:LRIP1^{Sic}(Tib)* were used as controls. **m** *CYP71A13* expression in various single substitution lines at 24 h post-inoculation with BCE. *LRIP1^{Tib}(R684/L876)* represents the substitution of Hap-Tib AAs with Hap-Sic AAs at specific positions in the *LRIP1^{Tib}* coding region. In **b-e**, **h-k**, **m**, Col, *LRIP1^{Sic}OE*, and *LRIP1^{Tib}OE* were used as controls. *LRIP1^{Sic}8m* represents the substitution of 8 Hap-Sic AAs with Hap-Tib AAs in the *LRIP1^{Sic}* coding region, while *LRIP1^{Tib}8m* represents the substitution of 8 Hap-Tib AAs with Hap-Sic AAs in the *LRIP1^{Tib}* coding region at the specific positions indicated in (**a**). In **b-m**, all data are presented as means $\pm$ s.d. ($n = 3$ biological replicates). Statistically significant differences ($P < 0.05$) were determined using one-way ANOVA with Tukey's multiple comparisons test and are indicated by different letters. Source data are provided as a Source data file.

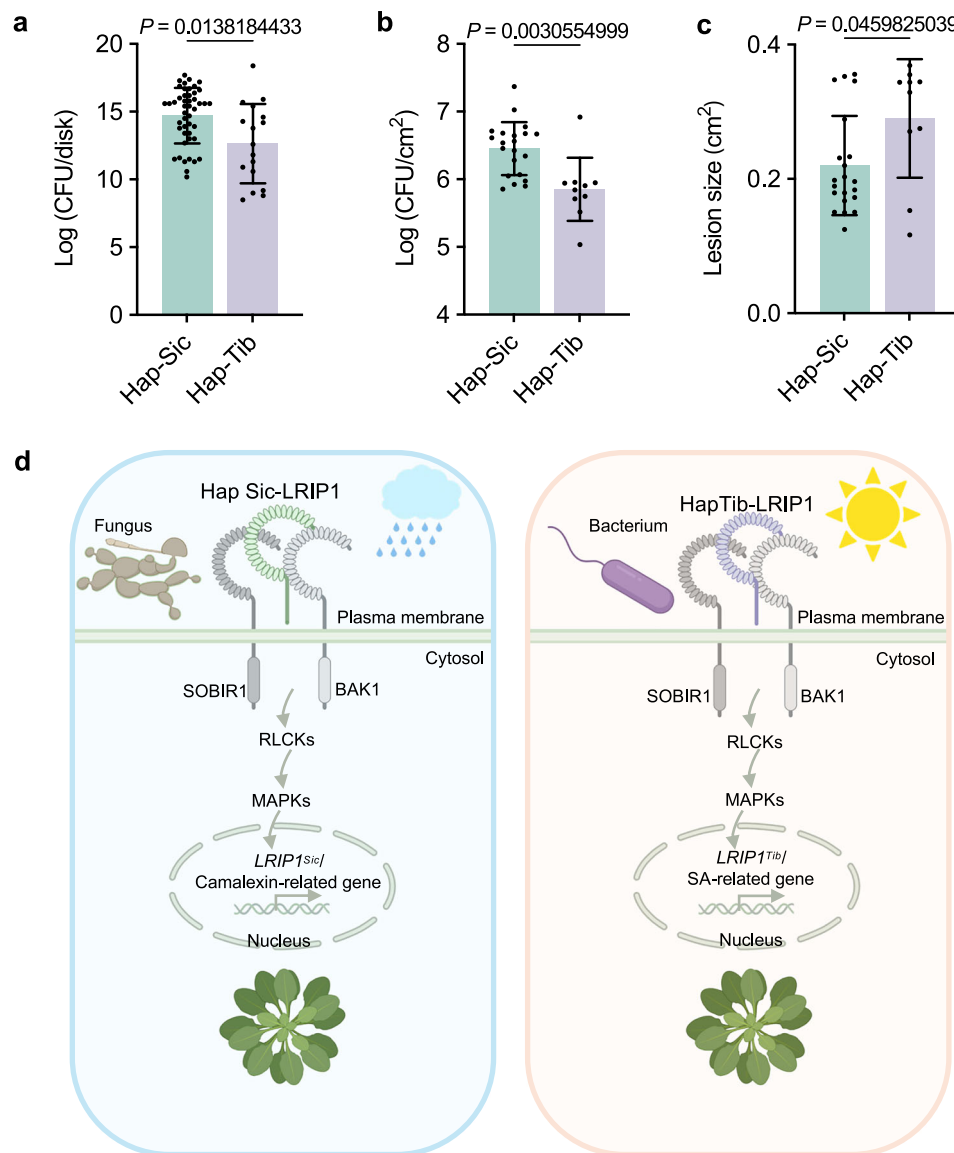


Fig. 7 | Phenotypic analysis of *Pst* DC3000 and *B. cinerea* resistance in different *A. thaliana* haplotype groups. **a** Phenotypic data on responses to *Pst* DC3000 from 64 accessions were obtained from previously published studies⁵⁵. Bacterial growth of *Pst* DC3000 (**b**) and lesion size following *B. cinerea* infection (**c**) were measured in 31 selected accessions, grouped into Hap-Sic and Hap-Tib based on phylogenetic analysis. Nine biological replicates were collected per accession, and the mean value of each accession was used for statistical analysis and displayed in the figure. In **a–c**, phenotypic differences between Hap-Sic and Hap-Tib accessions are presented as means $\pm$ s.d. ($n = 47$ Hap-Sic and 17 Hap-Tib accessions in **a**); $n = 21$ Hap-Sic and 10 Hap-Tib accessions in **b, c**). Statistical significance was assessed using an unpaired two-tailed t-test with Welch's correction, and exact P values are indicated. Source data are provided as a Source data file. **d** The proposed schematic model of LRIP1^{Sic} and LRIP1^{Tib}-mediated immune signaling in *Arabidopsis* across different ecosystems. Created in BioRender. LIU, B. (2026) <https://BioRender.com/>

id1stjk. LRIP1 functions as a non-canonical LRR immune protein associated with the plasma membrane through interaction with the co-receptor SOBIR1. Upon pathogen-derived elicitor stimulation, LRIP1 recruits BAK1 in an elicitor-dependent manner, leading to activation of downstream signaling cascades involving receptor-like cytoplasmic kinases (RLCKs) and MAPKs. Allelic variation in LRIP1 confers signaling specificity. LRIP1^{Sic} preferentially mediates responses to fungal-derived elicitors, promoting camalexin-related gene expression, whereas LRIP1^{Tib} preferentially mediates responses to bacterial-derived elicitors, activating SA-associated defense pathways. Promoter polymorphisms regulate inducible expression of *LRIP1* under different pathogen conditions, while CDS variation determines downstream signaling outputs. Together, these features define a modular system in which transcriptional regulation and protein variation jointly shape pathogen-specific immune responses.

similarities to well-characterized LRR-containing immune proteins, such as RLP23 and RLP42, are consistent with a role for LRIP1 in cell-surface immune signaling pathways, likely mediated through dynamic association with the co-receptors BAK1 and SOBIR1^{18–21}. Notably, we uncover allelic variation in LRIP1 across *A. thaliana* ecotypes that modulates pathogen-specific immune responses (Fig. 7d), suggesting a degree of flexibility beyond the classical JA–SA trade-off.

Functional dissection of two LRIP1 alleles revealed distinct immune outputs: the LRIP1^{Sic} allele mediates responses to BCE,

whereas LRIP1^{Tib} preferentially responds to D36E, each triggering overlapping yet distinct transcriptional and signaling profiles (Figs. 2, 3). Despite these differences, both alleles require SOBIR1 and recruit BAK1 in an elicitor-dependent manner, indicating that co-receptor dynamics may underlie signaling specificity. The combination of plasma membrane localization and the absence of a predicted transmembrane domain suggests that LRIP1 may associate with the membrane indirectly, potentially through interaction with membrane-anchored co-receptors. Together, our findings support a model in

which allelic variation in LRPI1 reprograms immune signaling outputs through differential co-receptor engagement, highlighting how diversification of receptor function at the allelic level may contribute to adaptation to diverse pathogen pressures. Similar strategies are observed in rice, where different allelic R proteins interact with distinct avirulences to optimize resistance⁴², highlighting the importance of host-pathogen coevolution for immune optimization. Further parallels can be drawn from liverworts, where two homologs, LYK1 and LYK2, evolve differentially to discern symbiotic and pathogenic microbes⁴³. However, unlike LYK1/2, LRPI1 allelic variants within a single locus in *A. thaliana* exemplify antagonistic pleiotropy, where enhanced resistance to one class of pathogens coincides with susceptibility to another—a balancing act shaped by evolutionary pressures. This trade-off underscores how plants optimize receptor function to reconcile competing immune demands without relying on hormonal cross-regulation.

Future studies should address how LRPI1 allelic complexes channel signals into divergent pathways. Immunoblot data implicates post-translational modifications in modulating LRPI1 activity, potentially via allele-specific interactions with other RLKs (Supplementary Fig. 6d and 7c). Identifying these modifications and their effector kinases through extensive biochemical and structural analyses coupled with large-scale screening will clarify the molecular basis of such signaling specificities. Moreover, harnessing LRPI1 allelic diversity could inform crop breeding strategies—introgression of LRPI1^{Tib} or LRPI1^{Sic} alleles into agronomic species may enhance resistance to regionally prevalent both types of pathogens. Collectively, our findings elucidate how allelic variation in a single LRR immune protein enables adaptive immune specialization, offering insights into evolutionary trade-offs and actionable targets for engineering pathogen-resilient crops.

Methods

Plant materials and growth conditions

Transgenic *A. thaliana* plants overexpressing LRPI1 (LRPI1^{OE}), LRPI1^{Sic} (LRPI1^{Sic}OE), and LRPI1^{Tib} (LRPI1^{Tib}OE) were generated by amplifying the CDS of LRPI1, LRPI1^{Sic}, and LRPI1^{Tib} from genomic DNA of the Col, Sic, and Tib ecotypes, respectively. The amplified fragments were cloned into the pCAMBIA1300-MYC vector using XbaI and KpnI restriction sites with the ClonExpress II One Step Cloning Kit (C112-01, Vazyme, Nanjing, China). For the construction of promoter-gene fusions (pLRPI1:LRPI1, pLRPI1^{Sic}:LRPI1^{Sic}, pLRPI1^{Tib}:LRPI1^{Tib}, pLRPI1^{Tib}:LRPI1^{Sic}, and pLRPI1^{Tib}:LRPI1^{Tib}), 2-kb promoter fragments of each allele were fused with the corresponding full-length CDS and cloned into the pBIB-BASTA-35S-GWR-GFP vector via KpnI and BamHI sites. For the pLRPI1^{Sic}:GUS and pLRPI1^{Tib}:GUS constructs, 2-kb promoter regions of each allele were fused to the CDS of GUS gene and subsequently cloned into the pCXGUS-P vector through the XcmI restriction site. Transgenic plants were screened with glufosinate ammonium or hygromycin, and transgene expression was validated by qRT-PCR. Reciprocal transformation lines, pLRPI1^{Tib}:LRPI1^{Tib}(Sic) and pLRPI1^{Sic}:LRPI1^{Sic}(Tib), were identified by sequencing (Sangon Biotech). The constructs were introduced into *A. thaliana* ecotypes Col, Sic, and Tib through the floral dip method⁴⁴. T-DNA insertion mutants, including SALK_094846 (*lrp1-1*) and SAIL_224_D05 (*lrp1-2*), were obtained from the Arabidopsis Biological Resource Center (ABRC), while the *bak1-5 bkk1*, and *sobir1-12* mutant seeds were kindly provided by Dr. Jianming Zhou and Dr. Yuelin Zhang, respectively. The *lrp1*^{Sic} mutants were generated using the CRISPR-Cas9 system based on a previously described method⁴⁵. To generate pLRPI1:LRPI1 *lrp1* and pLRPI1^{Sic}:LRPI1^{Sic} *lrp1*^{Sic} plants, genetic crosses were carried out and genotyped by genomic PCR. Primer sequences used for cloning are provided in Supplementary Data 6, and a complete list of transgenic lines is available in Supplementary Table S1. The F3 populations from confirmed lines were used for subsequent analyses. Unless otherwise

specified, all experiments were conducted in the Col background. Seeds were surface-sterilized in 10% sodium hypochlorite for 10 min, rinsed six times with sterile distilled water, and sown on half-strength Murashige and Skoog medium (Sigma-Aldrich, St. Louis, MO, USA) supplemented with 2% (w/v) sucrose and solidified with 0.7% (w/v) agar (pH 5.8). Plates were incubated at 22 °C under a 16 h light/8 h dark photoperiod with fluorescent white light (137 lux; Philips F17T8/TL841 17W). At the two-leaf stage, seedlings were transplanted into soil and maintained under the same growth conditions for subsequent experiments.

Inoculation analysis, PAMP treatments, and BCE preparation

For all *Pst* DC3000 and D36E inoculation assays, the 7th–9th true leaves from *A. thaliana* plants at a comparable developmental stage were selected. Bacterial suspensions (OD₆₀₀ = 0.002) were prepared in 10 mM MgCl₂ and pressure-infiltrated into the abaxial side of the leaves using a needleless syringe. Following inoculation, plants were maintained at 22 °C under high humidity to promote infection. Bacterial growth was quantified by harvesting 0.4 cm² leaf discs at designated time points. Discs were homogenized in 10 mM MgCl₂, and the resulting homogenates were serially diluted and plated onto King's B medium supplemented with rifampicin (50 mg/L) for CFU enumeration.

For *B. cinerea* inoculation, fungal spores were cultured on potato dextrose agar plates at 28 °C for 3–5 days until robust mycelial growth and sporulation were observed. Agar plugs (~5 mm in diameter) containing actively growing mycelium were excised and placed mycelium-side down onto *A. thaliana* leaves that had been lightly wounded using a sterile needle. Inoculated plants were then maintained under dark and high-humidity conditions to promote fungal infection.

For canonical PAMP treatments, 7th–9th rosette leaves were excised and incubated overnight in liquid MS medium prior to treatment. The following PAMPs were applied at the indicated concentrations: flg22 (100 nM; synthetic peptide, GenScript, cat. no. RP19986), elf18 (100 nM; synthetic peptide, GenScript), nlp20 (1 μM; synthetic peptide, GenScript), and chitin (100 μg/mL; Sigma-Aldrich, cat. no. C7170).

To prepare BCE (*B. cinerea* elicitor), mycelium was cultured in half-strength potato dextrose broth at 22 °C with shaking at 150 rpm for 5 days. The culture medium was filtered through Miracloth (Millipore) to remove fungal biomass, followed by centrifugation at 10,000 × g for 20 min at 4 °C to clear residual debris. The soluble fraction was concentrated and desalted using PD-10 desalting columns (Sigma-Aldrich, Cat# GE17-0851-01). Crude elicitor preparations were standardized according to the quantity of input material, calculated as the mass of lyophilized mycelium per mL of extract. BCE was applied to detached leaves incubated in liquid MS medium under the same conditions as described for PAMP treatments. All experiments were performed with at least three independent biological replicates, with a minimum of 20 leaves per replicate.

Genome-wide association analysis

We conducted GWAS using an F7 RIL population derived from a cross between the Sic and Tib ecotypes. A total of 120 RILs were phenotyped under controlled conditions for two traits: bacterial growth following *Pst* DC3000 inoculation and lesion size after *B. cinerea* infection (Supplementary Data 1).

For SNP identification, resequencing reads from the 120 RILs were aligned against the parental reference genome with BWA (v0.7.17). PCR duplicates were flagged using Sambamba (v0.7.1), and SNPs were identified with GATK (v4.1.1.7). Low-quality variants were removed with VCFtools (v0.1.15), retaining only SNPs with minor allele frequency ≥5% and missing rate ≤20%.

Association analyses were performed with GEMMA (v0.98.1) based on a univariate mixed linear model. To reduce false positives, a

kinship matrix was included to account for relatedness among lines. Genome-wide significance thresholds were determined using Bonferroni correction ($P < 0.05/n$, where n is the number of SNPs tested). A prominent association peak was detected on chromosome 2 for both phenotypes, overlapping with the *LRIP1* locus (*AT2G34930*). Haplotypes were grouped according to sequence variation in this region, and their mean phenotypic values (bacterial growth and lesion size) were compared to assess resistance differences.

RNA sequencing and transcriptome analysis

The 7th–9th leaves were used for all assays. D36E inoculation and BCE treatments were performed as described above. Total RNA was extracted using TRIzol reagent (BioFit; cat. no. RN33). Libraries were prepared and sequenced on an Illumina platform. Raw reads were trimmed by Trimmomatic⁴⁶. Clean reads were aligned to the *A. thaliana* reference genome (TAIR10)⁴⁷ using HISAT2⁴⁸, with SAMtools⁴⁹ employed for data sorting and file format conversion. Transcript assembly and read counting were performed using a union-exon approach with StringTie⁵⁰ restricted to the reference annotation. Transcripts per kilobase million (TPM) fragments mapped were calculated for each gene. Raw read count matrices were subsequently generated using the prepDE.py script. Differentially expressed genes were identified using the variance stabilizing transformation method in DESeq2 with the thresholds of $|\log_2(\text{fold change})| \geq 1$ and adjusted p -value < 0.05 (Supplementary Data 5)⁵¹.

qRT-PCR analysis

Total RNA was extracted from rosette leaves of 3-week-old *A. thaliana* plants using TRIzol reagent (BioFit, Bowling Green, OH, USA; Cat# RN33) following the manufacturer's instructions. 1 µg of purified RNA was used for cDNA synthesis with the PrimeScript RT reagent kit (Takara, Kyoto, Japan; Cat# RR047A). Gene expression was normalized to *EIF4A*, which served as the internal control. Quantitative real-time PCR (qRT-PCR) was carried out using the QuantiNova SYBR Green PCR Kit (Qiagen) and gene-specific primers listed in Supplemental Data 6 on a Bio-Rad CFX96 Real-Time PCR Detection System. Relative expression levels were determined following the method described previously⁵².

GUS staining

Plants grown under identical environmental conditions were subjected to either *Pst* DC3000 or *B. cinerea* treatment prior to histochemical analysis. *Pst* DC3000 inoculation was performed as described above, while *B. cinerea* infection was achieved by spraying a conidial suspension (5×10^5 spores/mL) uniformly onto the leaf surface. For GUS staining, leaf samples were incubated in 50 mM GUS staining buffer (Solarbio, Beijing, China; Cat# G3061) at 37 °C for 3 h. Following incubation, tissues were immersed in absolute ethanol overnight to remove chlorophyll and excess stain before imaging.

Co-IP

MYC-tagged LRIP1 protein variants were extracted from the 7th to 9th rosette leaves of 3-week-old *A. thaliana* transgenic lines (*LRIP1OE*, *LRIP1^{Sic}OE*, and *LRIP1^{Tib}OE*). Following infiltration treatments, the leaves were immediately flash-frozen in liquid nitrogen. Frozen tissues were ground into a fine powder, and total proteins were extracted using pre-chilled immunoprecipitation (IP) buffer (50 mM Tris-HCl, 150 mM NaCl, 1 mM EDTA, 1% Triton X-100, pH 7.5) supplemented with 1× protease inhibitor cocktail (PMSF). The homogenates were centrifuged at $12,000 \times g$ for 15 min at 4 °C to remove cellular debris and insoluble material. The resulting supernatants were incubated with pre-washed agarose-conjugated monoclonal anti-MYC antibody beads (Sigma-Aldrich, Cat# SAB5900007) equilibrated in IP buffer at 4 °C for 4 h with gentle rotation. To minimize nonspecific binding, the beads were washed three times with IP wash

buffer (20 mM HEPES, 40 mM KCl, 1% Triton X-100). Bound proteins were eluted by boiling the beads in SDS loading buffer at 95 °C for 10 min. The antibodies used for immunoblotting were anti-MYC (Sigma-Aldrich; Cat# M4439; dilution: 1:5000), anti-BAK1 (Agrisera; Cat# AS12 1858; dilution: 1:5000) and anti-SOBIR1 (Agrisera; Cat# AS16 3204; dilution: 1:1000).

Protein isolation and Western blot analysis of MPKs

Total protein extracts were prepared from the 7th to 9th rosette leaves of 3-week-old *A. thaliana* plants treated with either D36E ($OD_{600} = 0.02$) or BCE (100 µg/mL). Samples were collected at the indicated time points post-infiltration. Protein samples were prepared by pooling leaves from six distinct transgenic lines. Extraction was performed as previously described, and equal protein amounts were analyzed by SDS-PAGE and Western blotting. Anti-p-p44/42 MAPK antibody (also known as anti-PERK1/2 antibody; Cell Signaling Technology; Cat# 4370) was used to detect phosphorylated MPKs at a dilution of 1:2000.

ROS detection assays

ROS production was measured using a luminol-based chemiluminescence assay with minor modifications from a previously published method⁵³. Leaf discs (4.5 mm diameter) were harvested from fully expanded 7th to 9th rosette leaves of 3-week-old *A. thaliana* plants and floated overnight in 200 µL sterile water in a 96-well plate. The next day, the water was replaced with luminol reaction buffer (30 mg/L luminol and 20 mg/L horseradish peroxidase; Sigma Aldrich) supplemented with BCE (100 µg/mL) or D36E (1×10^7 CFU/mL) as indicated. Luminescence was recorded at 45-s intervals for 60 min using a Varioskan Flash microplate reader (Thermo Fisher Scientific).

Transient expression and subcellular localization assays

Agrobacterium tumefaciens strain GV3101 carrying the expression vector was cultured to an OD_{600} exceeding 2.0, harvested by centrifugation, and resuspended in infiltration buffer containing 10 mM MgCl₂ and 10 mM MES (pH 5.7) to a final OD_{600} of 0.08 for subcellular localization analysis. *Nicotiana benthamiana* leaves were infiltrated with the bacterial suspension and maintained in darkness for 24 h, followed by 48 h under light. Fluorescence imaging was then performed using a Leica SP8 confocal laser scanning microscope.

Luciferase complementation imaging (LCI) assay

LRIP1^{Sic} or *LRIP1^{Tib}* CDSs were cloned into pCambia1300-nLUC vectors, and *BAK1* and *SOBIR1* were cloned into pCambia1300-cLUC vectors, respectively. Constructs were co-infiltrated into 4-week-old *N. benthamiana* leaves using *Agrobacterium* GV3101. Leaves were kept in darkness for 24 h and then under light for 48 h. D36E or BCE was infiltrated into the leaves, 1 mM luciferin was applied immediately thereafter, and luminescence was recorded 30 min post infiltration using a NightSHADE LB985 system.

Haplotype network construction and phylogenetic analysis

LRIP1 sequences were extracted from 737 *A. thaliana* accessions using variant data from the 1001 Genomes Project and a previously published dataset³⁰. Based on the Col reference annotation, the genomic region corresponding to the *LRIP1* locus was retrieved, and proteins for each accession were reconstructed from variant data. Multiple sequence alignment of *LRIP1* proteins was performed using MUSCLE implemented in MEGA X with default parameters (Supplementary Data 2). A maximum likelihood phylogenetic tree was constructed using IQtree v2.1.2⁵⁴ under the best-fit substitution model with 1000 ultrafast bootstrap replicates. Orthologous *LRIP1* sequences from representative *Brassicaceae* species (including *Arabidopsis lyrata*, *A.*

halleri, and *Capsella rubella*) were included to root the phylogeny (Supplementary Data 3). Nucleotide variants within the LRR domain were extracted and used to construct a statistical parsimony haplotype network using Network 10 with the median-joining algorithm (Supplementary Data 4).

Statistical analysis

Statistical significance was assessed using one-way or two-way ANOVA followed by Tukey's or Šidák's multiple comparisons test, or using two-tailed Unpaired t-test with Welch's correction (GraphPad Prism 10.5.0). Data are presented as mean $\pm$ s.d. from three independent biological replicates unless otherwise indicated.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The *LRIPI* raw sequencing dataset for 705 accessions was obtained from the Arabidopsis 1001 Genomes Project [<http://signal.salk.edu/atg1001/3.0/gebrowser.php>] and supplemented with 32 additional accessions from a previous study³⁰ (Supplementary Data 2). Phenotypic data on responses to *Pst* DC3000 were obtained from a previous study⁵⁵. Annual precipitation data were downloaded from WorldClim (www.worldclim.org/)⁵⁶; and the aridity index was obtained from the CGIAR-CSI GeoPortal [<https://cgiarcsi.community/data/global-aridity-and-pet-database/>] (Supplementary Data 4). The raw RNA-seq data generated in this study have been deposited in the China National Center for Bioinformation under accession number [PRJCA065230](https://www.ncbi.nlm.nih.gov/PRJCA065230). The genetic materials used in this study are available from the corresponding authors upon request. Source data are provided with this paper.

References

- Teixeira, P. J. P., Colaïanni, N. R., Fitzpatrick, C. R. & Dangl, J. L. Beyond pathogens: microbiota interactions with the plant immune system. *Curr. Opin. Microbiol.* **49**, 7–17 (2019).
- Spoel, S. H. & Dong, X. How do plants achieve immunity? Defence without specialized immune cells. *Nat. Rev. Immunol.* **12**, 89–100 (2012).
- Kourelis, J. & van der Hoorn, R. A. L. Defended to the nines: 25 years of resistance gene cloning identifies nine mechanisms for R protein function. *Plant Cell* **30**, 285–299 (2018).
- Weyer, A.-L. V. et al. A species-wide inventory of NLR genes and alleles in *Arabidopsis thaliana*. *Cell* **178**, 1260–1272 (2019).
- Dodds, P. N. & Rathjen, J. P. Plant immunity: towards an integrated view of plant–pathogen interactions. *Nat. Rev. Genet.* **11**, 539–548 (2010).
- Macho, A. P. & Zipfel, C. Plant PRRs and the activation of innate immune signaling. *Mol. Cell* **54**, 263–272 (2014).
- Li, L., Yu, Y., Zhou, Z. & Zhou, J.-M. Plant pattern-recognition receptors controlling innate immunity. *Sci. China Life Sci.* **59**, 878–888 (2016).
- Zhou, J.-M. & Zhang, Y. Plant immunity: danger perception and signaling. *Cell* **181**, 978–989 (2020).
- Mondragón-Palomino, M., Stam, R., John-Arputharaj, A. & Dresselhaus, T. Diversification of defensins and NLRs in *Arabidopsis* species by different evolutionary mechanisms. *BMC Evol. Biol.* **17**, 255 (2017).
- Yuan, M. et al. Pattern-recognition receptors are required for NLR-mediated plant immunity. *Nature* **592**, 105–109 (2021).
- Ngou, B. P. M., Ahn, H.-K., Ding, P. & Jones, J. D. G. Mutual potentiation of plant immunity by cell-surface and intracellular receptors. *Nature* **592**, 110–115 (2021).
- Lee, D. et al. Regulation of reactive oxygen species during plant immunity through phosphorylation and ubiquitination of RBOHD. *Nat. Commun.* **11**, 1838 (2020).
- Kong, L. et al. Dual phosphorylation of DGK5-mediated PA burst regulates ROS in plant immunity. *Cell* **187**, 609–623 (2024).
- Pruitt, R. N. et al. The EDS1-PAD4-ADR1 node mediates *Arabidopsis* pattern-triggered immunity. *Nature* **598**, 495–499 (2021).
- Helft, L. et al. LRR conservation mapping to predict functional sites within protein leucine-rich repeat domains. *PLoS ONE* **6**, e21614 (2011).
- Zipfel, C. et al. Bacterial disease resistance in *Arabidopsis* through flagellin perception. *Nature* **428**, 764–767 (2004).
- Zipfel, C. et al. Perception of the bacterial PAMP EF-Tu by the receptor EFR restricts Agrobacterium-mediated transformation. *Cell* **125**, 749–760 (2006).
- Bi, G. et al. *Arabidopsis thaliana* receptor-like protein AtRLP23 associates with the receptor-like kinase AtSOBIR1. *Plant Signal. Behav.* **9**, e27937 (2014).
- Albert, I. et al. An RLP23–SOBIR1–BAK1 complex mediates NLP-triggered immunity. *Nat. Plants* **1**, 1–9 (2015).
- Fan, L. et al. Genotyping-by-sequencing-based identification of *Arabidopsis* pattern recognition receptor RLP32 recognizing proteobacterial translation initiation factor IF1. *Nat. Commun.* **13**, 1294 (2022).
- Zhang, L. et al. Distinct immune sensor systems for fungal endopolygalacturonases in closely related Brassicaceae. *Nat. Plants* **7**, 1254–1263 (2021).
- Tameling, W. I. L. et al. Mutations in the NB-ARC domain of I-2 that impair ATP hydrolysis cause autoactivation. *Plant Physiol.* **140**, 1233–1245 (2006).
- Wang, J. et al. Reconstitution and structure of a plant NLR resistosome conferring immunity. *Science* **364**, eaav5870 (2019).
- Zhu, L. et al. A conserved extracellular LRR-only protein disrupts BAK1-mediated immune receptor formation to negatively regulate plant immunity. *New Phytol.* **250**, 1183–1199 (2026).
- Jacques, A. et al. NtLRP1, a tobacco leucine-rich repeat gene with a possible role as a modulator of the hypersensitive response. *Mol. Plant Microbe Interact.* **19**, 747–757 (2006).
- Jung, H. W. & Hwang, B. K. The leucine-rich repeat (LRR) protein, CaLRR1, interacts with the hypersensitive induced reaction (HIR) protein, CaHIR1, and suppresses cell death induced by the CaHIR1 protein. *Mol. Plant Pathol.* **8**, 503–514 (2007).
- Chen, J.-B. et al. An LRR-only protein promotes NLP-triggered cell death and disease susceptibility by facilitating oligomerization of NLP in *Arabidopsis*. *New Phytol.* **232**, 1808–1822 (2021).
- Caddell, D. F., Park, C.-J., Thomas, N. C., Canlas, P. E. & Ronald, P. C. Silencing of the rice gene LRR1 compromises rice Xa21 transcript accumulation and XA21-mediated immunity. *Rice* **10**, 23 (2017).
- Zhou, L. et al. Rice hypersensitive induced reaction protein 1 (OsHIR1) associates with plasma membrane and triggers hypersensitive cell death. *BMC Plant Biol.* **10**, 290 (2010).
- Kang, M. et al. The pan-genome and local adaptation of *Arabidopsis thaliana*. *Nat. Commun.* **14**, 6259 (2023).
- Zipfel, C. & Oldroyd, G. E. D. Plant signalling in symbiosis and immunity. *Nature* **543**, 328–336 (2017).
- Müller, T. et al. Plant infection by the necrotrophic fungus *Botrytis* requires actin-dependent generation of high invasive turgor pressure. *New Phytol.* **244**, 192–201 (2024).
- Chen, J. et al. Reprogramming and remodeling: transcriptional and epigenetic regulation of salicylic acid-mediated plant defense. *J. Exp. Bot.* **71**, 5256–5268 (2020).
- Liu, Y. et al. Diverse roles of the salicylic acid receptors NPR1 and NPR3/NPR4 in plant immunity. *Plant Cell* **32**, 4002–4016 (2020).

35. Nguyen, N. H. et al. Camalexin accumulation as a component of plant immunity during interactions with pathogens and beneficial microbes. *Planta* **255**, 116 (2022).
36. Zhou, J. et al. Multilayered synergistic regulation of phytoalexin biosynthesis by ethylene, jasmonate, and MAPK signaling pathways in *Arabidopsis*. *Plant Cell* **34**, 3066–3087 (2022).
37. Schuhegger, R. et al. CYP71B15 (PAD3) catalyzes the final step in camalexin biosynthesis. *Plant Physiol.* **141**, 1248–1254 (2006).
38. Nafisi, M. et al. *Arabidopsis* cytochrome P450 monooxygenase 71A13 catalyzes the conversion of indole-3-acetaldoxime in camalexin synthesis. *Plant Cell* **19**, 2039–2052 (2007).
39. Hok, S. et al. An *Arabidopsis* (malectin-like) leucine-rich repeat receptor-like kinase contributes to downy mildew disease. *Plant Cell Environ.* **34**, 1944–1957 (2011).
40. Ramonell, K. et al. Loss-of-function mutations in chitin responsive genes show increased susceptibility to the powdery mildew pathogen *Erysiphe cichoracearum*. *Plant Physiol.* **138**, 1027–1036 (2005).
41. Stellmach, H., Hose, R., Rade, A., Marillonnet, S. & Hause, B. A new set of golden-gate-based organelle marker plasmids for colocalization studies in plants. *Plants* **11**, 2620 (2022).
42. Kanzaki, H. et al. Arms race co-evolution of *Magnaporthe oryzae* AVR-Pik and rice Pik genes driven by their physical interactions. *Plant J.* **72**, 894–907 (2012).
43. Tan, X. et al. A pair of LysM receptors mediates symbiosis and immunity discrimination in *Marchantia*. *Cell* **188**, 1330–1348 (2025).
44. Clough, S. J. & Bent, A. F. Floral dip: a simplified method of *Arabidopsis thaliana* transformation. *Plant J.* **16**, 735–743 (1998).
45. Lowder, L. G. et al. A CRISPR/Cas9 toolbox for multiplexed plant genome editing and transcriptional regulation. *Plant Physiol.* **169**, 971–985 (2015).
46. Bolger, A. M., Lohse, M. & Usadel, B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* **30**, 2114–2120 (2014).
47. Lamesch, P. et al. The *Arabidopsis* Information Resource (TAIR): improved gene annotation and new tools. *Nucleic Acids Res.* **40**, D1202–D1210 (2012).
48. Kim, D., Paggi, J. M., Park, C., Bennett, C. & Salzberg, S. L. Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. *Nat. Biotechnol.* **37**, 907–915 (2019).
49. Li, H. et al. The sequence alignment/map format and SAMtools. *Bioinformatics* **25**, 2078–2079 (2009).
50. Pertea, M. et al. StringTie enables improved reconstruction of a transcriptome from RNA-seq reads. *Nat. Biotechnol.* **33**, 290–295 (2015).
51. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 550 (2014).
52. Miura, K. et al. SIZ1-mediated sumoylation of ICE1 controls CBF3/DREB1A expression and freezing tolerance in *Arabidopsis*. *Plant Cell* **19**, 1403–1414 (2007).
53. Kadota, Y. et al. Direct regulation of the NADPH oxidase RBOHD by the PRR-associated kinase BIK1 during plant immunity. *Mol. Cell* **54**, 43–55 (2014).
54. Nguyen, L. T. et al. IQ-TREE: a fast and effective stochastic algorithm for estimating maximum-likelihood phylogenies. *Mol. Biol. Evol.* **32**, 268–274 (2015).
55. Atwell, S. et al. Genome-wide association study of 107 phenotypes in *Arabidopsis thaliana* inbred lines. *Nature* **465**, 627–631 (2010).
56. Fick, S. E. & Hijmans, R. J. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* **37**, 4302–4315 (2017).

Acknowledgements

We thank Dr. Jianming Zhou (Institute of Genetics and Developmental Biology, Chinese Academy of Sciences) and Dr. Yuelin Zhang (College of Life Sciences, Sichuan University) for kindly providing the *bak1-5 bkk1*, and *sobir1-12* mutant seeds, respectively. The proposed schematic model was created with BioRender.com. This research was supported by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (JYB2025XDXM910), the Science and Technology Major Project of Xizang (XZ202402ZD0005), National Natural Science Foundation of China (Grant Nos. 32030006, 31870244, and 31670317), and the Gansu Provincial Top-leading Talents Program to J.L. This work is also supported by the Big Data Computing Platform for Western Ecological Environment and Regional Development and the Supercomputing Center of Lanzhou University.

Author contributions

Conceptualization: B.L., Y.J., and J.L. Methodology: B.L., Y.Z., S.L., and J.H. Investigation: B.L., Y.Z., S.L., J.H., X.F., L.F., Y.S., and D.Z. Data curation: B.L. and Y.Z. Formal analysis: B.L., S.L., and J.H. Visualization: B.L. and Y.Z. Writing-original draft: B.L. and Y.Z. Writing-review & editing: B.L., Y.Z., Y. Zha, Y.J., and J.L. Supervision: Y. Zha, Y.J., and J.L. Funding acquisition: Y.J. and J.L.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains Supplementary material available at <https://doi.org/10.1038/s41467-026-75012-w>.

Correspondence and requests for materials should be addressed to Yuanzhong Jiang or Jianquan Liu.

Peer review information *Nature Communications* thanks Zhu-Qing Shao, who co-reviewed with Sai-Xi Li and the other anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026